

ARTIFICIAL IMPOSIBILITY

How Finance, Institutions, Biology, and Culture Make Feasible Futures Unbuildable



**AN OUTFINITIST MAP FOR SCIENCE FICTION, PUBLIC
DEBATE, AND THE AGE OF SCALABLE INTELLIGENCE**

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*How Finance, Institutions, Biology, and Culture
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Author's Note

This work was created under the umbrella of Axiologic Research as part of two interconnected research programs, one dedicated to Artificial Intelligence and the other to Outfinitism, both led by Sînică Alboaie. To explore the details of how these two programs intersect and build upon each other, we invite readers to visit the Outfinity Venture Studio page at www.outfinity.ch.

While Artificial Intelligence language models were used to assist with text generation, the foundational philosophy, thematic structure, and analytical approach derive exclusively from the author's decades of dedicated study of social phenomena, social technologies, and genuine hard technologies resulting from various European and self funded research projects. Recently, within the Achilles research project, we have been deeply studying AI generated literature and its automated verification using neuro symbolic approaches; all the free books made available to the public are part of the suite of works we use to test our latest technologies.

From a Science-Fiction Cabinet to an Outfinitist Map

This book began as a cabinet of premises for science fiction. The original question was not which technologies are physically impossible, but which useful projects a technologically advanced society might know how to build and still fail to produce. The early catalogue included public artificial intelligence, preventive medicine that destroys its own market, scientific institutions without conventional reviewers, platforms that help users consume less, radically personalized education, collective ownership of robots, machine fiduciaries, digital minds with rights, and international systems designed to prevent races that every participant privately fears. Each premise could support a novel. Each also seemed to describe a serious political problem arriving ahead of schedule.

The catalogue grew, and then became less illuminating. The surface variety concealed a smaller number of recurring structures. An antibiotic is unlike an open scientific protocol, and both are unlike a treaty on autonomous weapons, yet all can fail because the institution that would need to support them is rewarded for behavior incompatible with their success. A universal tutor and a public compute facility are different technologies, yet both can create value too diffusely to support conventional capture. Replacing a human reviewer and granting rights to an artificial mind are morally distant questions, yet both encounter a responsibility vacuum: a capability becomes plausible before society has identified an actor with legitimate authority to use it.

Once these recurrences became visible, the project changed. It was no longer primarily a list of speculative inventions. It became an inquiry into artificial impossibility: the condition in which a future is blocked not by physics but by the interaction of finance, law, administrative categories, institutional mandates, strategic rivalry, biological competition, inherited beliefs, and the need for legitimate authority. The word artificial does not mean imaginary. A human-made limit can be harder than a natural obstacle. A company without a payer closes; a treatment without a trial remains unavailable; a grant proposal outside an eligible category is never evaluated; a public agency without authority cannot act; a government that restrains itself unilaterally may expose its citizens to rivals. These constraints are real because institutions make them real.

Nor does artificial impossibility imply conspiracy. Most of the cases in this book require no hidden coordination and no villain. Venture investors can be competent fiduciaries, reviewers conscientious scientists, regulators defenders of rights, managers responsible stewards, voters reasonable parents, and states prudent guardians of security. The impossible future emerges from composition. Each actor follows a local mandate; the mandates do not add up to a sponsor for the whole. Local rationality produces global absence.

This makes the subject part of the Outfinitist philosophy adopted here. Outfinitism begins from finite but movable limits. It rejects both the fantasy that intelligence or resources will become literally boundless and the fatalism that treats every inherited boundary as a law of nature. A limit can be constructed, historically contingent, and still costly to move. Moving it can reveal another limit behind it. The aim is therefore not to proclaim that everything is possible. It is to distinguish physical

impossibility from institutional refusal, and protective friction from obsolete obstruction, so that societies can decide which boundaries deserve preservation and which have merely escaped examination.

Artificial intelligence makes this distinction urgent. Scalable intelligence can lower the cost of analysis, coordination, design, coding, literature review, monitoring, and persuasion. It can make projects feasible before their ownership, liability, financing, or legitimacy has been designed. It can also strengthen the very systems that block them: investors can screen faster, incumbents can lobby more effectively, grant applicants can generate more polished conformity, and bureaucracies can automate old categories. Technological acceleration does not automatically produce institutional acceleration. It often widens the gap.

Science fiction remains an essential use of the map. The most consequential speculative worlds are not defined only by gadgets. They reveal who owns the infrastructure, who receives the dividend, whose refusal matters, which institutions survive after their declared function becomes cheap, and what forms of status remain scarce after material abundance. A story can function as an institutional simulation. It can ask not merely whether a technology works, but what selection system admits it, what constituency opposes it, what moral category it violates, and which hidden job of an existing institution it accidentally destroys.

The same map belongs in public debate. Many of the projects examined here should not be built in their strongest form. Some barriers protect privacy, pluralism, due process, safety, or the right not to be optimized. The point of the book is not to convert technical feasibility into political command. It is to make the refusal intelligible. A society that can name the pattern blocking a project can debate the

pattern rather than arguing vaguely about whether the project is realistic. It can ask whether a Missing Sponsor needs a public mandate, whether a Success Penalty needs a different payment model, whether a Responsibility Vacuum requires new law, or whether a Sacred Boundary should remain intact.

The manuscript combines three kinds of argument. Where empirical research exists, it is used: studies of venture decision-making, R&D grants, research peer review, status competition, public goods, organizational legitimacy, and language-model sycophancy. Where direct evidence is weak, the book makes institutional inferences from known incentives. Where both evidence and precedent are absent, it labels the idea as a speculative scenario. The map is therefore not a catalogue of predictions. It is a vocabulary for asking better questions about futures that are technically near, socially valuable, and institutionally homeless.

A Note on the Argument

This is not an argument against markets, venture capital, peer review, regulation, religion, or liberal democracy. Each solves genuine coordination and legitimacy problems. The narrower claim is that every selection system has a domain of competence and a characteristic shadow. Artificial impossibility appears in that shadow, especially when several systems reject the same project for different but locally rational reasons. Evidence and institutional examples were reviewed through August 2026. The concept of smoothing is used as a working research term developed by Axiologic Research and connected here to established literatures on institutional decoupling, legitimacy, system justification, and AI sycophancy.

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INTRODUCTION

The Map of Artificial Impossibility

1. When Feasibility Is Not Enough

For most of history, impossibility named a frontier of nature. Humans could not fly, communicate instantly across oceans, transplant organs, or search a planetary library in seconds. Progress meant moving that frontier through theories, instruments, energy, organization, and engineering. The decisive question was whether a thing could be made to work.

The twenty-first century adds another frontier. A growing class of projects can be prototyped, and sometimes demonstrated, without finding a stable route into society. The technology is not magic. The social value is not absurd. Yet the project cannot identify a payer, satisfy the return profile of available capital, fit a grant category, survive procurement, obtain liability coverage, coordinate the necessary actors, or acquire legitimate authority. The project is feasible in engineering and homeless in institutions.

Consider automated scientific review. Language models and retrieval systems can already perform parts of literature comparison, consistency checking, methodological critique, and novelty assessment. By 2026, end-to-end and multi-agent systems had demonstrated increasingly broad portions of scientific workflows, while studies of AI review showed both useful signal and serious weaknesses, including weak agreement with experts and vulnerability to adversarial text (Lu et al., 2026; Ghareeb et al., 2026; Sandström and Thelwall, 2026; Lin et al., 2025). It is therefore plausible that much repetitive review labor will

become automatable. But reviewers do more than detect errors. They exercise disciplinary judgment, compare opportunity costs, interpret strategic priorities, and lend accountable human authority to decisions. The technology can arrive before society has decomposed the role.

Or consider an AI system designed to reduce consumption. It compares products, exposes hidden costs, recommends repair, advises the user not to buy, and ends the interaction when no purchase is needed. Such a system may increase autonomy and reduce waste. It is also hostile to advertising, affiliate revenue, engagement metrics, and many forms of platform growth. Its most faithful behavior destroys its easiest business model.

A third example is prevention. A system that prevents disease, cyberattack, flood damage, fraud, or administrative error creates value through events that do not happen. The beneficiaries are distributed, counterfactual, and often located in another budget or another decade. The organization that pays may not be the organization that saves. In medicine, preventive services have often been expected to save money while treatments are assessed more conventionally through benefits and risks, an asymmetry noted by Pryor and Volpp (2018). In antibiotics, the mismatch is even sharper: society needs innovation but should avoid maximizing sales. The United Kingdom's subscription approach and WHO advocacy for pull incentives are attempts to pay for availability rather than volume (UK Government, 2024; WHO, 2024).

These are not isolated market failures. A society does not evaluate a project from one point of view. Markets ask whether customers pay. Venture funds ask whether a small ownership stake can become disproportionately valuable within a finite fund life. Grant systems ask whether a proposal is legible, comparable, eligible, and defensible.

Procurement asks whether requirements can be specified and awarded fairly. Regulators ask who bears risk and liability. Democratic politics asks which coalition will authorize the transition. Families, professions, religions, and status systems ask what identities and hierarchies the project disturbs. Geopolitics asks whether restraint creates vulnerability.

Artificial impossibility is the result of these selection systems acting together. The definition used in this book has five parts. The project must be technically plausible enough to investigate rather than pure fantasy. It must offer substantial potential human value. Ordinary actors following current mandates must repeatedly lack a rational path to sponsor it. Translating it into an accepted startup, grant, product, or regulatory category must damage a central mechanism of its value. Finally, a credible change in financing, ownership, law, or governance must be able to move the boundary. If no conceivable institutional change matters because the underlying science is impossible, the case belongs to a different book.

Artificial does not mean unjustified. Some limits are deliberate achievements. Due process makes automated decisions slower because speed is not the only public value. Privacy restricts useful data flows because people are not raw material. Democratic authorization prevents technically superior planners from acquiring unlimited power. The purpose of the map is to locate the decision, not to predetermine it. A named barrier can be defended, redesigned, compensated, experimentally relaxed, or rejected. An unnamed barrier is mistaken for reality itself.

2. Fifteen Patterns of Artificial Impossibility

A pattern language is more useful than a catalogue because it compresses without pretending that cases are identical. The names below are deliberately intuitive. They are not proposed as settled scientific constructs. They are diagnostic handles: compact ways to notice that an antibiotic, a public AI system, and a humane social network may fail for homologous reasons even though their technologies, users, and moral stakes differ.

Pattern	Meaning and rapid examples
The Missing Sponsor	The benefit is real but no actor is authorized and rewarded to pay for it. Examples: cross-agency evidence infrastructure, biodiversity archives, or universal scientific tools.
The Power-Law Orphan	The project may become useful and sustainable but cannot plausibly return a venture fund. Examples: low-margin accessibility technology, local infrastructure, or slow deep research.
The Success Penalty	Success reduces future revenue or requires restrained use. Examples: prevention, cures, antibiotics, durable products, and organizations designed to dissolve after completing a mission.
The Commons Penalty	The socially valuable form is open, interoperable, or neutral, while finance seeks enclosure and control. Examples: public compute, standards, open models, and provenance networks.
The Hostile Patron	The actor best positioned to fund the project loses revenue, authority, status, or assets if it works. Examples: reviewer automation, credential alternatives, or anti-lock-in protocols.
The Proof Trap	Evidence is required before permission or funding, but the evidence can be produced only after permission or funding. Examples: rare-disease trials, radical research, and new public institutions.
The Legibility Tax	The project crosses categories, changes its problem definition, or evolves faster than calls, budgets, procurement, and regulation can follow.
The Time-Horizon Rift	Costs are immediate and concentrated; benefits arrive slowly, elsewhere, or after the sponsor has changed. Prevention, maintenance, climate adaptation, and foundational research often fit.
The Coordination Threshold	Value appears only after many actors adopt a standard, pool data, or share risk. The first mover cannot capture enough benefit to justify leading.

Pattern	Meaning and rapid examples
The Responsibility Vacuum	Capability exists, but no legitimate actor will own the error, coercion, or irreversible consequence. Automated review, medical AI, and law simulation encounter this gap.
Veto Asymmetry	Benefits are diffuse while losses are concentrated among organized incumbents. The project has a social surplus but no coalition willing to own the transition.
The Positional Ceiling	Technology can multiply goods but cannot make everyone top-ranked, exclusive, admired, or desired. Abundance collides with status, prestige, location, and mating competition.
The Restraint Race	All actors benefit if everyone limits a dangerous capability, yet unilateral restraint transfers advantage to less restrained rivals.
The Sacred Boundary	A feasible project crosses constitutive ideas about personhood, family, autonomy, merit, religion, nature, or what may legitimately be bought and optimized.
The Smoothing Veil	Institutions, cultures, and AI systems preserve a coherent story by omitting losers, external costs, temporal effects, contradictions, or alternatives that do not fit the accepted frame.

Most difficult projects combine patterns. Public scientific compute can be a Missing Sponsor because benefits cross agencies, a Commons Penalty because interoperability matters, a Power-Law Orphan because the operating organization may be durable rather than explosive, and a Responsibility Vacuum because allocation rules influence who can conduct frontier research. A universal comparison agent can carry a Success Penalty by reducing purchases, a Hostile Patron because platforms and retailers benefit from the opacity it removes, and a Smoothing Veil because consumer choice is usually discussed without exposing the persuasion systems that shape it.

The patterns also interact causally. The Proof Trap increases the Legibility Tax: when evidence is scarce, administrators demand more plans and credentials. The Time-Horizon Rift strengthens Veto Asymmetry: beneficiaries who gain later organize less intensely than incumbents who lose now. The Commons Penalty creates a Missing Sponsor: once value is intentionally allowed to spill outward, no owner can capture enough of it. The Responsibility Vacuum encourages smoothing because institutions describe a normative choice as a technical problem until a legitimate decision maker can be found.

Naming the pattern changes the question. Instead of asking why investors fail to understand a project, one can ask whether it is a Power-Law Orphan that should never have been presented to venture capital. Instead of asking why reviewers are conservative, one can ask how much of the decision requires prediction, how much requires legitimacy, and where the Proof Trap makes honest uncertainty appear incompetent. Instead of treating cultural resistance as ignorance, one can ask whether the project crosses a Sacred Boundary that cannot be moved by more data alone.

The table therefore does not abolish detail. It disciplines it. Each chapter returns to a small number of patterns and examines their mechanisms, evidence, and failure modes. The examples are no longer the argument. They are tests of whether the abstraction illuminates more than it hides.

3. How the Map Was Built

The unit of analysis in this book is not a technology by itself. It is a project inside a selection environment. The same capability can be

buildable under one ownership and impossible under another. A private language model sold to enterprises may fit venture capital; a neutral model that every public institution can audit and use at marginal cost may require infrastructure finance and a public constitution. A review agent sold as productivity software may have customers; an agent authorized to allocate public grants requires legitimacy, appeal, and a different liability regime.

The method begins by reconstructing the project without forcing it into an existing organizational form. What outcome creates the value? Who benefits, including people who never become customers? Which counterfactual event is avoided? Which assets must remain open, portable, or neutral? Which rights must not be traded for revenue? What time horizon matters? This step often reveals that the startup pitch, grant application, or policy label has already smoothed away the core difficulty.

The next step is to identify the selecting institutions and their legitimate mandates. A venture capitalist is not asked to maximize public welfare; a pension trustee is not free to gamble beneficiaries' savings on any beautiful idea; a regulator cannot waive rights merely because innovation is attractive; a reviewer cannot confidently certify knowledge that does not exist. Critique becomes serious only when it respects these constraints. Otherwise it degenerates into the claim that everyone should behave like a benevolent philosopher with unlimited capital and no accountability.

The third step is the counterfactual test. Would a plausible change in payment, ownership, procurement, liability, or governance make the project buildable without destroying the values that justified it? The test prevents the category from becoming a romantic label for every

rejected idea. Some projects lack evidence because they are weak. Some markets do not exist because users do not care. Some technologies are forbidden because the risks are unacceptable. Artificial impossibility is most useful when the barrier is specific enough to redesign.

A fourth step is to examine transition politics. Every institutional improvement redistributes rents, roles, prestige, and risk. A system that reduces grant-writing bureaucracy also reduces the value of grant-writing expertise. Competency certification may widen access while weakening universities' signaling power and cross-subsidies. Automated prevention may save a health system while reducing revenue for organizations built around treatment. A proposal that describes aggregate benefit without describing organized loss is incomplete.

Finally, the analysis performs a smoothing audit. Smoothing, in the sense used by Axiologic Research, is the production of a more coherent and manageable representation by suppressing relations that make the picture harder to accept or optimize. It is broader than factual error. A statement can be locally true and globally smoothing: "venture capital finances innovation," "peer review selects excellent science," "AI saves time," or "consumers choose freely." Each sentence names a real function while hiding selection effects, opportunity costs, intensification, or the machinery that constructs choice.

Smoothing is not always harmful. Institutions need categories; models need compression; people need narratives that permit action. A description containing every causal connection would be unusable. The danger appears when the omitted relation would change the decision, when the same stakeholder repeatedly disappears, or when a temporary simplification hardens into the ontology of the system. The book

therefore asks, throughout: which actor vanishes from the story, which cost moves to another budget or decade, which metric substitutes for the real objective, which failure is rewritten as progress, and which conflict is presented as a mere implementation gap?

This method produces a map rather than a verdict. The map can guide research, institutional design, and fiction. It can also reveal that some limits should remain. The central Outfinitist commitment is not boundless construction but conscious boundary management: know which limit is natural, which is social, which is protective, which is extractive, and what new limit will appear when the old one moves.

Evidence discipline

Kind of claim	How it is treated in this book
Empirical finding	A claim supported by quantitative or qualitative research with a method that can be inspected. The text cites the relevant study and avoids extending it beyond its domain.
Institutional evidence	A program, law, funding mechanism, or organizational practice that demonstrates a problem or attempted solution, without proving general causal effects.
Institutional inference	A logical conclusion derived from mandates, ownership, cash flows, liability, or strategic incentives. It is presented as an argument, not as an established empirical law.
Speculative scenario	A plausible extrapolation used for science fiction or anticipatory debate. Its value lies in exposing a structure before evidence can exist.

CHAPTER ONE

Capital Cannot Want Everything

Modern culture often uses venture capital as a synonym for technological seriousness. This chapter treats it instead as one highly effective selection machine with a precise return logic. The important question is not whether venture capital is good or bad, but which futures become distorted when every ambitious project is required to resemble a venture-backed company.

1. The Power-Law Gate

Venture capital is often described as money for innovation. That description is correct in the same way that a turbine is a machine for producing motion: it omits the precise conditions under which the machine works. Venture capital is not patient admiration for promising ideas. It is a specialized institutional technology for financing a portfolio of highly uncertain firms when conventional lending is impossible, information is asymmetric, assets are difficult to collateralize, and a small number of successes must compensate for many failures.

The structure matters. Venture funds are generally organized as finite-life private partnerships. Limited partners commit capital; general partners select companies, stage investments, influence governance, and seek liquidity through acquisitions, secondary sales, or public markets. The underlying assets are illiquid, and investors accept long lockups. Academic surveys describe a typical private-equity fund life of roughly

ten years, often with extensions, and empirical work on venture decision-making shows the importance of team quality, market size, business model, product, and exit prospects (Metrick and Yasuda, 2011; Gompers et al., 2020). A fund is not paid in social benefit. It must return cash or liquid securities to its limited partners.

This creates a selection function. An investable company must not merely be useful. It must possess a credible route to becoming disproportionately valuable within the time and ownership structure of the fund. Because many investments fail and many survivors produce modest outcomes, a fund depends on a minority of companies capable of returning a large fraction of the portfolio. The exact statistical shape of venture returns is debated and varies across datasets, sectors, and stages, but the practical logic is widely recognized: the upside of a successful investment must be large enough that missing it can matter more than avoiding several ordinary failures.

The resulting questions are familiar. Can this become a very large market? Can revenue grow much faster than costs? Can the company defend its margin? Can it raise subsequent rounds? Is there a plausible buyer or public market? Can ownership remain valuable even if competitors imitate the underlying technology? Does the team appear capable of recruiting, selling, and surviving the next financing stage? These questions are not cynical. They are how a venture fund remains solvent.

Yet they silently redefine innovation. A project that improves life for millions can be a poor venture investment if each person benefits only slightly, if public agencies are the natural customer, if the technology should be open, if adoption requires common standards, if the main benefit is avoided harm, or if the likely outcome is a durable

medium-sized organization rather than an extreme exit. Conversely, a project can be an excellent venture investment while producing ambiguous social value if it creates strong lock-in, captures attention, extracts rents from an information asymmetry, or converts a public practice into a proprietary transaction layer.

The distinction is between value creation and value capture. Venture finance cannot rely on the first alone. A company must capture enough of the value it creates to support rapid growth and a high terminal valuation. Many of the most socially important systems are designed to let value escape: standards, protocols, scientific knowledge, preventive practices, open-source components, public-health surveillance, interoperable identity, environmental monitoring, and educational resources. Their purpose is to improve the productivity and freedom of many downstream actors. The more successfully they do so, the more difficult it may be for the originating organization to own the resulting gains.

Time creates another filter. A project may have an enormous expected value but require fifteen or twenty years of scientific, regulatory, and industrial development. Deep-technology companies often face long and uncertain R&D cycles, expensive prototypes, manufacturing scale-up, and two distinct valleys of death: from research to marketable product and from initial product to industrial scale. European policy documents now describe these problems explicitly, and the European Innovation Council has adopted blended finance and patient-capital language partly because ordinary private capital often does not match the risk and duration of frontier technologies (European Commission, 2025a; EIC, 2026a; OECD, 2025a).

Even “patient” venture capital is patient relative to public markets, not relative to civilization. A fund can hold an asset for years, but its clock is still finite. The technology must become financeable at each stage. A company that needs a decade before the next credible valuation event may fail even when the ultimate science is sound. This produces a peculiar kind of epistemic selection: projects are favored when progress can be narrated in financing-compatible increments. A visible prototype, a growing user count, a regulatory milestone, or a partnership can support the next round. Slow accumulation of negative knowledge—learning which hypotheses fail, building reliable datasets, discovering edge cases, or maintaining long-term observations—often cannot.

The staging of capital also changes organizational behavior. Each round is both financing and examination. Founders are encouraged to present a future large enough to justify dilution and risk. Moderation can be interpreted as lack of ambition. A technically appropriate plan to build a stable, interoperable infrastructure may be rewritten as a platform strategy with dominant-market potential. A project whose social value depends on neutrality may be pushed toward enclosure because neutrality is difficult to value as equity. A privacy technology may be encouraged to monetize data-adjacent services. A scientific tool may be advised to become the proprietary system of record. The project is not necessarily corrupted by bad people. It is translated into a form the capital structure can recognize.

The problem becomes sharper in AI. Software prototypes can be built quickly, and narratives of rapid scale are plausible. This increases competition for attention and rewards demonstrations that compress ambiguity into a compelling story. But the hardest AI

problems—reliability, verification, provenance, adversarial robustness, governance, long-term evaluation, domain integration, public infrastructure—may require slow work whose benefits are distributed. The venture filter can overproduce visible capability and underproduce the institutions that make capability trustworthy.

This does not imply that venture capital should be reformed into philanthropy. A vehicle that no longer seeks venture returns will no longer attract capital on venture terms. The more useful conclusion is classificatory: venture capital is one instrument with a precise domain of competence. It is excellent when a project can internalize value, scale rapidly, defend margins, and produce liquidity. It is structurally mismatched to projects whose success requires openness, restrained use, universal access, perpetual stewardship, politically contested authority, or benefits that accrue mainly outside the firm.

A society that treats the venture-backed startup as the default embodiment of technological progress will therefore misclassify a large portion of its future. It will interpret absence of investment as absence of value. It will ask public goods to become monopolies, infrastructures to become apps, scientific institutions to become SaaS products, and civilizational insurance to promise an exit. Some projects can survive this translation. Others become unbuildable precisely because the available money insists that they become something else.

2. The Anti-Venture Project

Some projects are merely unattractive to venture investors. Others are anti-venture in a stronger sense: the mechanism by which they create social value directly undermines the mechanism by which an

equity investor would earn an outsized return. Their purpose is to reduce consumption, eliminate dependency, distribute ownership, prevent events, expose information asymmetries, lower switching costs, or make a scarce capability abundant. They are not bad businesses accidentally. They are good social projects whose success weakens the ordinary business model.

The clearest example is the self-erasing market. Imagine a company founded to eliminate a disease, a form of fraud, an administrative burden, or a category of accident. If it succeeds completely, demand disappears. This can still be investable when the company can sell a cure, license a patent, or expand into adjacent markets. But the pure form is structurally unstable. The organization develops employees, investors, suppliers, and management whose livelihoods depend on persistence. Even without misconduct, it will search for new definitions of the problem, new customer segments, or recurring services. Institutions rarely dissolve when their founding problem is solved.

Antibiotic innovation reveals an especially perverse version. Society needs effective new drugs but should use them sparingly to preserve effectiveness. A sales-volume model rewards the opposite of stewardship. WHO and G7 discussions increasingly describe the need for pull incentives, subscription payments, and delinked models because the public-health value of a reserve antibiotic is not represented by units sold (WHO, 2024; WHO, 2025). Here success is not merely market-ending; it requires deliberately suppressing the market.

Prevention has a related structure. A preventive system creates value through events that do not occur. The avoided heart attack, breach, flood, relapse, failed experiment, or infrastructure collapse is counterfactual. It is difficult to prove which intervention caused the

absence, and the beneficiary may be different from the payer. A landlord pays for resilience that protects an insurer. An employer funds health interventions whose savings accrue to a future employer or national health service. A city invests in heat adaptation whose greatest benefit appears during a rare event after current officials have left office. Prevention can be economically valuable while remaining financially homeless.

The anti-venture pattern can be formalized through four mismatches.

The first is benefit leakage. Most of the value flows to actors who do not pay the producer. A standard improves an industry. A safety benchmark disciplines many vendors. A scientific dataset generates discoveries in other organizations. A clean protocol lowers transaction costs across a network. The project creates a positive externality by design.

The second is metric inversion. The available revenue metric moves opposite to the social objective. Engagement rises when addiction worsens. Treatment volume rises when prevention fails. replacement sales rise when products break. Security spending rises after incidents. Advertising revenue rises with persuasion intensity. A company can attempt to redefine the metric, but it competes against firms rewarded by the old one.

The third is ownership incompatibility. The project must remain credibly neutral, open, or collectively governed. A single owner with a right to sell, close, reprice, or discriminate would reduce adoption or undermine legitimacy. This applies to identity systems, public-interest data trusts, evaluation standards, election infrastructure, and some

research platforms. The very governance needed for social value limits the equity story.

The fourth is time-horizon dispersion. Costs are concentrated now; benefits accumulate slowly across institutions and generations. Climate adaptation, foundational education, biodiversity preservation, pandemic preparedness, and long-term scientific archives fit this structure. Even when net present social value is high, no private balance sheet captures enough of it soon enough.

Entrepreneurs often respond by adding a commercial wrapper. A public-good core becomes a premium platform. A prevention system sells analytics. An open model is supported by hosting. A neutral protocol gains a managed service. These hybrids can work and should not be dismissed. The danger is mission inversion: the commercial complement becomes the governing objective, and the public core is maintained only while it supports capture. Investors may rationally discourage interoperability, encourage data enclosure, or prioritize high-paying customers over universal access. The social infrastructure becomes an acquisition funnel.

The same tension appears in projects that would reduce monopoly rents. A universal comparison agent that reveals total product cost, durability, repairability, privacy consequences, and alternatives could improve market competition. But it would threaten advertising, branding premiums, opaque pricing, and platform commissions. A fiduciary AI legally bound to act for the user could recommend not buying, switching providers, negotiating aggressively, or leaving a platform. It would be exceptionally valuable to users and strategically hostile to many potential distributors.

A fully effective scientific review infrastructure could threaten journals, consultancies, grant-writing services, and some forms of academic gatekeeping. A legal agent that makes ordinary contracts comprehensible could reduce billable ambiguity. A public procurement assistant that detects overspecification and vendor lock-in could weaken incumbent suppliers. A system that automatically validates project milestones could reduce the role of reporting intermediaries. These are technologies of de-rentification: they create value by removing the informational or procedural friction from which existing organizations earn income.

This is why some apparently obvious public-interest projects remain permanently in pilot form. A philanthropic grant proves that the technology works. A small government program demonstrates value. An open-source community builds the core. But no institution finances universal deployment, long-term maintenance, governance, support, and adaptation. Venture investors want stronger capture. Public agencies want a procured vendor with liability and service guarantees. Philanthropy wants novelty and visible impact. Users want the system to be free. The project dies between proof and permanence.

The appropriate conclusion is not that anti-venture projects cannot involve companies. It is that their capital structure must recognize the inversion. Payment may need to be tied to verified social outcomes rather than volume, as in an advance market commitment. A public or cooperative entity may own the core while firms compete in complements. A perpetual trust may protect neutrality. A regulated utility model may permit stable returns without monopoly extraction. Procurement may guarantee demand before the market exists. A prize

may reward creation while requiring open licensing. A citizen fund may own automation infrastructure and distribute dividends.

Without such structures, the project is asked to perform an institutional contradiction: maximize private capture by reducing private capture; grow usage by reducing usage; preserve a problem by solving it; build a monopoly in order to create a commons. The contradiction is not resolved by a better pitch deck.

3. The Shape of the Claim

The usual response to the venture-capital filter is to invoke the diversity of finance. A project unsuitable for a venture fund can, in principle, seek bank debt, private equity, project finance, corporate investment, insurance capital, customer prepayments, impact capital, philanthropy, a family office, a cooperative, or the state. This is true and often practically useless. Finance is not generic money stored in differently branded containers. Every instrument recognizes a particular claim.

Equity recognizes a residual claim. It accepts uncertainty because the upside can be large, but it needs an ownable organization whose value can eventually become liquid. Debt recognizes a contractual claim. It does not require extreme upside, but it needs repayment visibility, collateral, guarantees, or stable cash flow. Project finance recognizes a bounded asset with allocated risks and contracted demand. Corporate capital recognizes strategic advantage: a technology should strengthen the parent, defend a market, reduce cost, or open an adjacent business. Insurance recognizes a reduction in losses that the insurer will actually bear. Customer finance recognizes a current user

able to commit. Philanthropy recognizes donor purpose. Public finance recognizes a lawful mandate and an accountable expenditure.

The artificial impossibility appears when the project's benefit cannot be translated into any of these claims without being changed. Foundational scientific knowledge is difficult to collateralize and should often be non-exclusive. Prevention may save a different institution from the one that pays. Public infrastructure may require neutrality that an acquirer could revoke. A humane platform may need to recommend departure, reducing the revenue that would service investment. A technology for future generations has beneficiaries who cannot sign contracts. The problem is not insufficient persuasion. The claim itself has no natural holder.

Research on R&D finance supports the underlying asymmetry. Hall and Lerner's review of the financing of research and development emphasizes the high cost of external capital for small and new innovative firms, driven by uncertainty, information problems, and the intangible character of knowledge. Venture capital mitigates part of this gap but does not remove it. Howell's quasi-experimental study of US Department of Energy SBIR grants found that an early-stage award approximately doubled the probability of later venture funding and increased patenting and revenue, especially for financially constrained firms (Hall and Lerner, 2010; Howell, 2017). Public money did not merely certify a winner; it financed the prototype that private investors needed to see.

This suggests that projects should be financed by risk stage rather than organizational identity. Epistemic risk belongs with grants, research endowments, and bounded exploratory awards. Engineering risk may fit milestone contracts, prizes, or patient equity. Demand risk

can be reduced through advance market commitments, subscriptions, or procurement. Operating infrastructure can use regulated returns, availability payments, membership fees, or public budgets. Long-term stewardship can sit in a trust, cooperative, foundation, or public institution after the period of entrepreneurial construction ends.

The resulting capital stack is not a compromise in which every financier receives a little of what it wants. It is a sequence of constitutional changes. Control and return rights should change as uncertainty falls. A founder may need freedom during exploration, investors may need influence during scale-up, users may need representation once the system becomes essential, and a neutral core may eventually need protection from sale. Ordinary corporate law can accommodate parts of this through benefit corporations, mission locks, capped returns, golden shares, perpetual-purpose trusts, steward ownership, cooperatives, and public-interest conditions. The difficulty is that these structures are less standardized, less liquid, and often misunderstood by investors and procurement systems.

Benevolent money does not eliminate the problem. Impact investing can widen sector choice, but a fund that promises market-rate returns still selects projects compatible with market-rate returns. Concessionary capital can accept lower returns, yet must measure impact and defend additionality. Philanthropy can fund unpopular science, rare diseases, open tools, and institutional experiments, but it often prefers pilots and catalytic grants to permanent operations. Leadership changes, donor priorities move, and successful pilots enter a graveyard of systems that proved useful without acquiring an operating institution.

Blended finance is more promising because it admits that risks have different owners. Public or philanthropic capital can absorb early uncertainty, guarantees can make infrastructure bankable, and private capital can contribute execution and scale. Yet the distribution of upside must be explicit. If the public absorbs the downside while investors receive unrestricted gains, the arrangement can become socialized risk and privatized reward. Access obligations, royalties, public equity, price conditions, open standards, and conversion to stewardship ownership are not decorative ethical clauses. They determine whether the financing preserves the project's reason for existing.

Corporate and insurance capital illustrate the Hostile Patron. A cloud provider may support portability until perfect portability weakens its advantage. A pharmaceutical company can invest in prevention while comparing it with recurring treatment revenue. An insurer may benefit from long-term health prevention but lose the customer before savings arrive. None of these actors needs to misunderstand the project. They may understand it precisely and still prefer a version that protects their balance sheet.

The same analysis applies to ownership. Venture culture imagines the technology company as a temporary capitalization table moving toward liquidity. Some systems need the opposite: credible continuity, neutrality, universal access, and protection against acquisition. A scientific provenance network, identity layer, safety benchmark, or public-interest AI service loses value if users believe a future owner can close, reprice, discriminate, or repurpose it. The project may need ownership without exit.

Ownership without exit does not forbid revenue or private participation. It separates financing from ultimate control. Investors can receive capped returns, priority payments, royalties, or redemption. Commercial firms can build services above an open core. A public buyer can finance construction while a trust protects the protocol. The organization can scale through federation rather than unified monopoly. What it cannot promise is that success will culminate in the unrestricted sale of the institution's constitution.

The practical lesson is that “try another investor” is not an innovation policy. The better questions are: which uncertainty is being financed, what claim can legitimately be owned, who receives the benefit, who bears the downside, when should control change, and what must remain outside the transaction? A project rejected by venture capital may be correctly rejected by venture capital and still be one of the most important projects society has failed to build.

CHAPTER TWO

The State Funds What It Can Legibly Defend

Private capital rejects projects it cannot own. Public funding rejects projects it cannot justify. The second filter is less visible because it speaks in the language of excellence, fairness, eligibility, and accountability. Its central artifact is the proposal: a document that must make uncertain knowledge appear administratively plausible before the work begins.

1. The Legibility Engine

Public research funding has a problem that no procedure can eliminate: it must choose among claims about knowledge that does not yet exist. Applicants know more about their own ideas than funders do, outcomes are uncertain, and success may become visible only years later. Some failures are evidence of poor execution; others are the normal cost of testing genuinely risky hypotheses. A democratic state cannot simply hand money to whoever sounds inspired. It needs a procedure that appears impartial, informed, and accountable.

Peer review is the dominant answer. Experts assess significance, novelty, method, team, feasibility, and impact. Panels compare proposals. Program officers apply budgets and policy priorities. The result is not merely a prediction about future science. It is a defensible public decision: money was allocated according to criteria by independent specialists rather than personal favor.

That legitimating role explains peer review's persistence despite its documented limitations. Reviews of grant evaluation have reported low or variable inter-rater agreement, weak ability to predict outcomes near funding thresholds, bias connected to applicant characteristics or reputation, and high costs for applicants and reviewers (Guthrie et al., 2018; Bendiscioli, 2019; Recio-Saucedo et al., 2022; Schweiger et al., 2024). The concern remains active: a 2025 registered-report protocol proposed analysing 134,991 reviews across four Norwegian funders to identify the determinants of inter-rater agreement (Hesselberg et al., 2025). As a protocol, it should not be read as evidence of results; it shows instead the scale and persistence of the reliability question. Other analyses find moderate relationships between review scores and later output, reminding us that peer review is neither pure noise nor a reliable oracle (Gallo et al., 2014; Li and Agha, 2015). The defensible conclusion is not that review is useless, but that its precision is often overstated.

The most consequential error may not be arbitrary ranking. It is selection for legibility. A proposal must describe in advance why the question matters, what will be done, which risks exist, which outputs will be delivered, and how impact will occur. This is reasonable for accountable spending. It also favors projects that can tell a coherent story before the difficult discovery begins.

Radical research is often incoherent at proposal time. The problem may be visible but the method unknown. The decisive experiment may emerge only after several failures. A new concept may lack the accepted vocabulary through which reviewers recognize significance. Interdisciplinary work may appear shallow to each discipline because its contribution lies in the connection. A proposal can be most honest

when it admits that the path is uncertain, yet that honesty can be scored as weak feasibility.

The grant proposal therefore becomes a literary technology for converting uncertainty into plausibility. Researchers learn to describe exploratory work as a sequence of controlled tasks. Unknowns become risks with mitigation plans. Dependencies become work packages. Serendipity becomes an impact pathway. The document is not necessarily deceptive; it is a boundary object between inquiry and administration. But the performance has consequences. Scientists choose projects that can survive narration, institutions reward teams experienced in grant language, and early-career researchers learn that epistemic courage must be wrapped in procedural confidence.

This is one reason high-risk programs are created as exceptions. The NIH High-Risk, High-Reward Research program explicitly seeks ideas that may be too early or risky for traditional peer review, while DARPA structures fixed-term, high-risk programs and short explorations intended to establish feasibility rapidly (NIH, 2026; DARPA, 2026). Their existence is evidence of institutional self-awareness. The ordinary process is optimized for a different balance of accountability and risk.

Peer review also privileges consensus in a domain where disagreement can be informative. A proposal that all reviewers find competent may be a safe investment. A proposal that one reviewer considers transformative and another considers misguided may contain greater epistemic variance. Conventional averaging often treats variance as uncertainty to be penalized. Portfolio logic might treat it as a signal of novelty worth a bounded experiment. Studies of disagreement in grant review have explored whether controversial applications produce

different outcomes, but funding systems rarely make disagreement itself a first-class design variable.

Reputation enters because reviewers face limited time and incomplete information. A famous investigator, institution, or method reduces uncertainty. Prior achievement is relevant evidence, but it can become cumulative advantage: resources flow to actors already positioned to produce credible proposals. This is the Matthew effect in institutional form. Blinding identities can reduce some cues, yet expertise and feasibility often depend on knowing the team. The challenge is not to eliminate prior information but to prevent it from silently dominating the evaluation of ideas.

Review burdens also matter. Scientists spend time writing and evaluating proposals rather than conducting research. When success rates fall, more effort is spent competing for a fixed pool, and the aggregate cost can become large even if each review is conscientious. Schweiger and colleagues characterize this as a cost of competition for scarce funds: applicant and reviewer labor can consume resources without increasing the pool of research performed (Schweiger et al., 2024). The problem is especially acute when proposals require elaborate plans that are rewritten after funding.

Publication peer review adds another filter. Journals assess completed manuscripts, but the decision still depends on novelty, perceived significance, presentation, and fit. AI-generated submissions and the growing volume of research increase reviewer pressure. By 2025–2026, surveys, specialized models, and benchmarks for automated scholarly review had moved from hypothetical discussion to active research. Systems can generate structured critiques, compare manuscripts with literature, and identify limitations; however, they

inherit historical review biases, show sensitivity to presentation, remain vulnerable to adversarial prompt injection, and can produce confident but ungrounded judgments (Zhuang et al., 2025; Idahl et al., 2025; Lin et al., 2025; Wu et al., 2026).

The temptation is to pose a binary question: should AI replace reviewers? This is premature because “reviewer” names several roles. Reviewers locate relevant prior work, check methods, identify unsupported claims, judge importance, interpret disciplinary norms, expose ethical problems, compare opportunity costs, and lend personal accountability to a decision. AI capability differs across these functions. Literature search and consistency checking may become highly automatable. The legitimacy to allocate public resources remains a separate matter.

There is also a danger of automating conservatism. A model trained on past decisions may become a high-speed replica of inherited taste. It can standardize the biases of prestigious venues, reward familiar rhetorical structures, and punish concepts that lack historical analogues. An automated reviewer optimized to agree with human reviewers is not necessarily optimized to identify valuable science. Agreement is a benchmark of imitation, not proof of epistemic superiority.

The opposite danger is romanticizing human intuition. Human reviewers are also biased, inconsistent, status-sensitive, and vulnerable to persuasive writing. They have limited time and cannot search the entire literature. The relevant comparison is not ideal human judgment against flawed AI, but hybrid institutional designs under real constraints. AI can make evidence visible, quantify disagreement, generate adversarial critiques, and expose missing controls. Humans

can define goals, interpret unusual contexts, accept responsibility, and protect plural conceptions of value.

A transformed funding system would treat review as a decomposable pipeline rather than a sacred ceremony. It would ask which parts are factual, which predictive, which normative, and which constitutional. It would measure reviewer calibration rather than assuming expertise. It would preserve minority arguments. It would recognize that proposals below the threshold may be statistically indistinguishable. It would use lotteries or bounded experiments where ranking precision is fictitious. It would evaluate portfolios, not only isolated proposals. Most importantly, it would stop demanding that genuine research promise not to surprise the funder.

2. Replacing Reviewers Without Creating a Sovereign Model

Replacing reviewers is an attractive slogan because review is slow, expensive, inconsistent, and often frustrating. It is also dangerously imprecise. A reviewer performs several distinct functions: verifying claims, evaluating methods, locating prior work, estimating feasibility, identifying risks, comparing proposals, interpreting strategic priorities, exercising expert judgment, and lending legitimacy to a decision. AI can transform these functions at different rates. Treating them as one indivisible role leads either to technological timidity or reckless automation.

The most automatable layer is clerical and analytical. Systems can check eligibility, budgets, internal consistency, citations, duplication, statistical assumptions, data-management plans, regulatory

dependencies, and the relation between objectives, activities, milestones, and evidence. They can retrieve relevant literature, identify contradictory findings, compare promised performance with known baselines, and test whether a proposal's novelty claims survive structured search. They can flag uncertainty rather than silently producing a score.

A second layer is adversarial evaluation. Multiple specialized agents can attack a proposal from different perspectives: scientific validity, engineering feasibility, security, privacy, ethics, market assumptions, public value, and implementation capacity. Proponents can respond, revise, and attach executable evidence. The result is closer to an open technical review than a sealed examination. AI is valuable here because critique can be repeated cheaply and because every claim can be connected to a trace.

A third layer is comparative judgment. Funding is scarce, so agencies must choose among proposals that may all be credible. Models can estimate portfolio overlap, thematic concentration, risk distribution, expected option value, and complementarity. They can reveal that ten individually strong proposals make a collectively conservative portfolio. They can simulate alternative allocations and show which assumptions drive the ranking. But comparison embeds value judgments: whether to prefer near-term impact, regional equity, scientific novelty, strategic autonomy, vulnerable populations, or maximum expected return.

A fourth layer is constitutional authority. Who has the right to spend public money, accept safety risk, or define a mission? This function cannot be inferred from predictive accuracy. Even a model that forecast outcomes perfectly would not possess democratic

authorization. Human institutions must determine objectives, rights, constraints, appeal procedures, and acceptable distributions of risk. AI can inform these decisions; it cannot confer legitimacy upon itself.

The evidence available by 2026 supports neither complacency nor replacement enthusiasm. Sandström and Thelwall found that open-weight language models' scores on a historical set of grant proposals correlated weakly but positively with expert scores, substantially below reviewer-to-reviewer agreement; they suggested triage or tie-breaking rather than autonomous allocation. Thorne and colleagues similarly found potential supplementary value but high variability and sensitivity to formal compliance. In publication review, Lin and colleagues showed that textual manipulations can distort LLM assessments. These findings are compatible with useful automation of bounded functions and incompatible with treating a fluent score as an epistemic verdict (Lin et al., 2025; Sandström and Thelwall, 2026; Thorne et al., 2026).

At the same time, the surrounding scientific workflow is changing quickly. Systems reported in Nature in 2026 performed increasingly broad chains of literature synthesis, hypothesis generation, experiment planning, implementation, result analysis, manuscript production, and critique (Asai et al., 2026; Lu et al., 2026; Gottweis et al., 2026; Ghareeb et al., 2026). The important implication is not that science has been automated. It is that the cost and speed of particular cognitive operations are becoming separable from the institutions that currently bundle them into expert roles.

An AI-native funding system should therefore replace documents before it replaces people. The conventional proposal is a static narrative written months before work begins. It compresses uncertainty into

confidence, separates claims from evidence, and becomes obsolete as the project learns. An AI-native proposal can be a living evidence graph: objectives, assumptions, dependencies, prior art, experiments, data, models, costs, risks, and decision points connected in machine-readable form. Narrative remains essential for meaning and communication, but it is no longer the sole executable object.

Review becomes continuous validation. At the start, the system evaluates whether an experiment is worth buying. During execution, it checks whether evidence supports continuation, revision, expansion, or termination. Periodic reports are generated from the trace rather than reconstructed as retrospective stories. Reviewers inspect anomalies, contested judgments, high-impact choices, and ethical questions instead of repeatedly verifying routine compliance.

This model supports smaller and faster grants. A researcher can request resources for the next uncertainty-reducing step rather than pretending to know a three-year path. The system can authorize a bounded amount under predefined rules, monitor outputs, and escalate when thresholds are met. Human program managers retain discretion but operate with richer evidence and a clear audit trail. The cost of applying must remain proportional; automation that invites thousands of elaborate machine-generated proposals without stronger screening would increase noise rather than access.

Automated novelty assessment is useful but should not become a universal oracle. Literature is incomplete, terminology varies, negative results are underreported, and genuinely new ideas can resemble old ideas at a superficial level. Models also inherit presentation biases and can be manipulated by prompt injection, strategic phrasing, or hidden text. Novelty systems should expose retrieved precedents and reasons,

allow rebuttal, and distinguish “not found” from “proven new.” Multiple retrieval and reasoning systems should be compared for high-stakes decisions.

Automated feasibility assessment has similar limits. Historical data favors projects resembling historical successes. Transformative research may look improbable precisely because it departs from the training distribution. A good system should not only predict success. It should identify which uncertainty could be resolved cheaply and what evidence would change the decision. In other words, it should fund information, not merely forecast outcomes.

Bias mitigation requires more than removing applicant names. Models can infer institutions, regions, disciplines, career stage, and writing culture from content. Standardized machine assistance may help applicants with weaker grant-writing support, but it may also homogenize style and reward those who optimize against known evaluators. Agencies should test systems adversarially across synthetic and historical cases, publish error analyses, and allow independent auditing. No model should silently define the linguistic style of fundability.

Transparency must be carefully scoped. Applicants deserve reasons and access to the evidence used against them. The public deserves aggregate information about allocations, conflicts, performance, and model governance. Full publication of every security-sensitive proposal, personal assessment, or model vulnerability may be harmful. The principle should be contestable opacity: secrecy requires a stated reason, an authorized custodian, an expiration or review point, and a route for challenge.

Appeal is not a cosmetic safeguard. Automated systems can make errors at scale and with an aura of consistency. Applicants need a way to identify incorrect sources, misunderstood methods, inaccessible evidence, conflicts of interest, and category errors. Appeals should be handled by evaluators who are not simply rerunning the same model. Successful appeals should update evaluation rules and become part of system monitoring.

Plurality is another safeguard. One model, benchmark, or optimization criterion should not control funding. Agencies can maintain independent evaluators with different architectures and mandates. Disagreement becomes information. A proposal accepted by all systems may be robust but conventional. A proposal that produces structured disagreement may deserve human attention or a small exploratory grant. Diversity in evaluation is valuable for the same reason diversity in research is valuable: correlated error is dangerous.

Ethical and constitutional review should be separated from technical scoring. A system can establish that a surveillance technology is feasible and effective without determining that it is legitimate. Rights, distributive effects, and irreversible social changes require publicly authorized principles. Citizen panels, affected communities, courts, regulators, and elected bodies may each have roles. The process will be slower than automated technical evaluation, and that slowness can be a safeguard.

The strongest version of AI-native funding is not an automatic grant vending machine. It is an executable institution. Its rules, evidence, models, decisions, and revisions are traceable. It can explain why a project was funded, what assumptions justified continuation, which failures were informative, and how the portfolio changed.

Independent researchers can evaluate the funder as rigorously as the funder evaluates applicants.

There is a danger that executable funding becomes excessively formal. Not every insight can be represented in a graph; not every transformative idea begins with clean evidence; not every social effect is measurable. Narrative, tacit knowledge, and trust remain necessary. The institution should use formalization to expose assumptions, not to deny whatever cannot yet be formalized.

The transition can begin incrementally. Agencies can automate eligibility and consistency checks, provide applicants with pre-submission critique, support reviewers with literature and evidence tools, publish structured decision rationales, and run parallel evaluations without delegating final authority. They can create experimental programs in which microgrants are allocated through staged evidence and compare outcomes with conventional review. They can fund independent red teams to attack the evaluation systems themselves.

Eventually, some routine reviews may disappear. Periodic reports can be generated automatically. Technical compliance can be continuously checked. Literature comparison can be machine-assisted by default. Human panels can become smaller, more specialized, and concentrated on contested judgments. This is a real partial replacement of reviewers, not merely administrative assistance. But the replaced labor is the labor of repetitive verification and document interpretation, not the democratic responsibility to decide what risks and futures a society should support.

The decisive principle is simple: automate evaluation where correctness can be tested; pluralize judgment where values conflict;

preserve human accountability where coercive authority is exercised. A funding system that follows this principle can become faster without becoming arbitrary, and more experimental without concealing failure.

AI will not remove politics from funding. It can remove much of the fog in which politics currently hides.

3. From Grants as Predictions to Funding as Experiments

Peer review receives most criticism because it visibly decides winners and losers. The deeper filter often appears earlier, in the architecture of the program. Before a reviewer reads a proposal, the funding institution has defined eligible organizations, thematic scope, expected outcomes, budget ranges, project duration, technology readiness, consortium composition, reporting obligations, and evaluation criteria. These choices determine which futures can be proposed in a recognizable form.

A public call is an attempt to translate political priorities into administrable competitions. It must be broad enough to permit innovation and narrow enough to justify expenditure. It must coordinate budgets approved in advance, legal rules, policy objectives, and implementation capacity. The call text therefore becomes a temporary constitution for a miniature research economy.

This architecture works best when the problem is understood and multiple teams can compete over solutions. It works less well when the decisive innovation changes the problem definition. A call for better document classification may exclude a proposal to redesign documents as executable knowledge. A program for trustworthy large language

models may struggle to recognize a neuro-symbolic architecture that rejects the language-model frame. A health call organized around diseases may miss infrastructure that transforms evidence integration across diseases. Institutions fund what they can name, while important research may be precisely the invention of a new name.

Timing is another filter. Calls open and close on fixed schedules. Consortia assemble months in advance. Evaluation can take months; the European Research Executive Agency notes that Horizon Europe proposal evaluation may last up to five months before grant preparation (REA, 2026). In fast-moving AI domains, a technical assumption can change between proposal drafting and project start. Teams must either freeze an outdated plan or reinterpret it after award. The more detailed the ex ante specification, the greater the divergence between contractual truth and scientific truth.

Large collaborative programs add consortium logic. Cross-border and cross-sector partnerships can combine expertise, create European or national added value, and distribute capacity. They can also produce consortia assembled for eligibility rather than epistemic necessity. Partners are assigned work packages because a balanced proposal needs roles. Coordination costs grow. Political and geographic considerations enter scientific architecture. Experienced organizations accumulate templates, networks, and administrative staff, while unconventional teams face a high entry cost.

The work-package structure encourages decomposition before understanding. Activities are divided into tasks, deliverables, milestones, dependencies, person-months, and risk tables. This is useful for engineering known systems. In exploratory science, it can become false precision. A research team may know that it needs to investigate a

phenomenon but not which conceptual route will survive contact with evidence. The administrative model treats changing direction as deviation, even when direction change is the most rational response to learning.

Reporting reinforces the problem. Periodic reviews ask whether promised work was completed, budgets used correctly, risks managed, and impact progressing. These controls protect taxpayers. They also reward projects that preserve the appearance of plan continuity. Failure becomes easier to hide through reinterpretation than to report as a valuable negative result. A project may produce documents demonstrating activity while the core hypothesis quietly dissolves.

The European Commission has recognized administrative burden and expanded lump-sum grants. Its 2025 interim evaluation estimated that lump sums could reduce beneficiaries' administrative costs by roughly 14 to 30 percent over a project's life by eliminating much financial reporting (European Commission, 2025b). This is meaningful reform. Yet simplified cost accounting does not by itself solve epistemic bureaucracy. A lump sum can still be attached to an over-specified work plan, and completion of a work package can remain a weak proxy for scientific gain.

Artificial intelligence changes the appropriate granularity of funding. If literature synthesis, coding, simulation, drafting, monitoring, and preliminary experimentation become faster, some research loops that once justified multi-year grants may be testable in months or weeks. The natural response is not simply to demand more outputs from the same grant. It is to reduce the size and duration of early commitments, increase the frequency of evidence-based

continuation decisions, and reserve large awards for stages where capital expenditure or deployment truly requires them.

Current systems often do the opposite. The transaction cost of applying is so high that grants must be large to be worth pursuing. Evaluation is expensive, so decisions are infrequent. Administrative capacity favors fewer large contracts. AI can lower the cost of producing proposals faster than institutions lower the cost of evaluating them, creating submission floods and declining success rates. The bureaucratic equilibrium becomes absurd: machines help applicants generate more elaborate documents, humans become more overloaded, and funders respond with stricter templates or automated filters that select polished compliance.

Public procurement is a second major bottleneck. Governments purchase enormous volumes of technology and services, so procurement could create markets for socially valuable innovation. In practice, procurement must protect fairness, competition, budget discipline, and legal defensibility. Buyers specify requirements, compare bids, verify qualifications, and avoid arbitrary favoritism. These safeguards can favor incumbents with certifications, balance sheets, references, and legal teams.

Innovation is difficult to specify because the buyer may not know the solution. Over-specification freezes old architectures. Under-specification creates evaluation risk. Procurement officers are often punished more visibly for a failed experiment than for purchasing a safe but inferior incumbent solution. European institutions now explicitly acknowledge legal and procedural barriers to innovation procurement and the underuse of procurement for startups and

scaleups (European Commission, 2025a; European Commission, 2026b).

The result is the pilot trap. A small innovative procurement or grant proves value under protected conditions. Scaling requires a new tender with requirements designed for reliability, integration, and liability. The original startup may lack the certifications or capacity. An incumbent wins and incorporates a diluted version. Alternatively, the pilot remains an isolated demonstration because no operating budget or procurement category exists for permanence.

Procurement also assumes a buyer. Some infrastructures benefit many agencies but belong to none. A cross-government evidence graph, reusable AI evaluation service, common data trust, or open agent standard may reduce costs across ministries while requiring one ministry to bear initial risk. Budget silos prevent investment in external benefits. The project needs a whole-of-government sponsor that ordinary departmental accounting does not provide.

Technology readiness levels create another simplification. They help distinguish basic research from prototypes and deployment. But a single scale can conceal multiple maturities. An algorithm may be technically mature while governance, data rights, user acceptance, and business models remain immature. Conversely, a social institution may be ready while a technical component evolves rapidly. Funding tied rigidly to TRL can move a project toward demonstration without resolving the institution around it.

Public funders also tend to separate research, innovation, regulation, and service delivery into different programs. Transformative projects cross these boundaries. A continuous health-prevention system is simultaneously medical research, digital infrastructure, insurance

reform, data governance, and public service. An executable-science platform is research infrastructure, publishing reform, AI development, standards work, and procurement. No call owns the whole. Applicants carve out a fundable slice, while integration—the source of much value—remains unfunded.

This is the bureaucracy of plausibility. Public systems do not reject the future because officials oppose it. They request that the future arrive as an eligible legal entity, within a named topic, at a specified maturity, through a balanced consortium, with a fixed duration, auditable costs, measurable impact, and manageable liability. Each requirement is defensible. Their intersection selects projects that already resemble the administrative past.

The danger is not bureaucracy itself. Liberal democracy needs records, appeals, non-discrimination, and constraints on discretion. The danger is confusing documentary compliance with accountability. AI and executable evidence could permit much stronger accountability with much less ceremonial reporting. A project might expose real-time provenance, experiments, expenditures, model versions, decisions, failures, and validation tests. The funder could inspect what happened rather than infer it from retrospective prose. Yet adopting such systems would reduce the importance of the document rituals by which institutions currently demonstrate control. Technical transparency can threaten administrative authority as much as it improves it.

The grant system is usually organized around a large prediction: a team describes what it will do over several years, reviewers estimate whether the plan is excellent and feasible, and the funder commits a block of resources. This architecture made sense when evaluation was expensive, experiments slow, reporting manual, and coordination

dependent on fixed documents. Scalable intelligence changes the relative costs. Literature synthesis, coding, simulation, drafting, monitoring, and preliminary testing can become faster while the institutional cycle remains measured in months or years.

The obvious response is often to demand more output from the same grant. The better response is to change the unit of commitment. Many early research questions can be funded as short, bounded uncertainty-reduction steps. A team receives enough resources to run a decisive experiment, build a benchmark, test a critical dependency, reproduce a claim, or falsify a premise. Evidence determines whether the next tranche is larger, different, or unnecessary. Large multi-year awards remain appropriate where laboratories, cohorts, infrastructure, fieldwork, or industrial scale require continuity. They cease to be the default shape of every serious idea.

This model is not speculation without precedent. Howell's analysis of SBIR grants suggests that early public funding can make technically uncertain firms legible to later investors by enabling prototyping. Azoulay, Graff Zivin, and Manso found that the Howard Hughes Medical Institute's investigator-oriented funding, which tolerated early failure and rewarded longer-term success, was associated with more exploratory and influential research than a more conventional funding comparison group (Azoulay et al., 2011; Howell, 2017). The NIH High-Risk, High-Reward program explicitly targets ideas that may be too early or risky to fare well in traditional review. These cases do not prove a universal design, but they show that funding rules shape the kind of uncertainty researchers can rationally accept.

An AI-native process could lower transaction costs enough to support a mixed portfolio. Routine eligibility, budget consistency,

citation verification, prior-art search, duplication checks, data requirements, and regulatory dependencies can be assessed continuously. Specialized agents can generate adversarial critiques from scientific, engineering, privacy, security, ethical, and implementation perspectives. Applicants can respond with evidence rather than another layer of rhetoric. Program managers can see disagreement, portfolio overlap, missing capabilities, and which assumptions control a decision.

The decisive object is no longer the static proposal. It is a living evidence structure connecting objectives, assumptions, sources, experiments, data, model versions, costs, risks, decisions, and failures. Narrative remains essential because meaning cannot be reduced to a graph, but narrative is no longer the sole record from which accountability must be inferred. Periodic review can be generated from the trace. A project that changes direction because evidence invalidates its plan need not pretend that the original work package was fulfilled through reinterpretation.

This matters for lump-sum funding. Simplified financial reporting reduces administrative burden, but it can still leave epistemic bureaucracy intact if completion is judged against an over-specified plan. Continuous evidence allows a stronger bargain: less ceremonial documentation in exchange for more transparent state. The funder can see what was attempted, why it changed, what failed, and what was learned. Failure becomes auditable without becoming shameful.

Replacing reviewers, in this architecture, means decomposing the role. Machines can increasingly replace repetitive verification, literature comparison, structural critique, and routine compliance. They can support comparative analysis, but should expose assumptions rather than silently rank. Human reviewers and program managers remain

responsible for unusual context, contested importance, ethical judgment, and portfolio choice. Elected institutions, courts, regulators, and affected communities retain authority over rights and coercive risk.

Qualified lotteries become sensible near thresholds where review cannot discriminate reliably. Randomness is not a denial of merit; it is an admission that several proposals may be good enough and that precise ranking can amplify noise, reputation, and writing advantage. A portfolio can combine conventional awards, investigator-centered support, microgrants, mission programs, replication budgets, and lottery allocation among qualified proposals. It can evaluate not only which projects won, but whether the selection system overconcentrates on familiar institutions, methods, regions, and narratives.

The greatest danger is not an AI that openly replaces reviewers. It is an AI that silently hardens the old review culture. If models are optimized to imitate historical scores, they can reproduce inherited taste at greater speed and scale. If applicants optimize against known evaluators, grant writing becomes an adversarial language game between machines. If the model rewards coherent plans, radical uncertainty becomes even less legible. An executable funding system must therefore preserve disagreement, permit appeal, compare independent evaluators, publish aggregate error analyses, and reserve resources for proposals that produce structured dissent.

The principle is straightforward: automate where correctness can be tested; pluralize where prediction is uncertain; preserve human accountability where values and public authority are involved. AI will not remove politics from funding. It can make the politics more visible and reduce the expensive prose in which it currently hides.

CHAPTER THREE

Value That Disappears When Monetized

Some projects are not merely difficult to monetize. Their social value is produced by reducing transactions, preventing events, opening foundations, or making the organization itself less necessary. These are the projects for which success is economically awkward.

1. The Success Penalty: Prevention and Self-Erasing Markets

The easiest project to finance is one with a visible buyer, a measurable product, and a revenue stream that rises with use. Many of humanity's most valuable projects have the opposite structure. They prevent an event, preserve an option, maintain a system, or remove the need for a recurring transaction. Their output is a non-event: the pandemic that did not spread, the bridge that did not collapse, the disease that did not progress, the cyberattack that did not succeed, the scientific error that did not propagate, the child who did not fall behind.

The economics of non-events are difficult because causality is counterfactual. After a successful prevention program, skeptics can argue that the feared event would not have occurred. After a failed program, the cost is visible. This asymmetry affects politics, insurance, public budgets, and private investment. Prevention competes against

immediate needs whose beneficiaries can point to present suffering. Maintenance competes against new construction that creates photographs, contracts, and political credit.

Health illustrates nearly every mismatch. A continuous preventive-health infrastructure could integrate biomarkers, behavior, environment, family history, and clinical evidence to identify risk trajectories early. AI could personalize recommendations and monitor whether interventions work. The potential benefit includes longer healthy life, lower treatment costs, and reduced burden on families. Yet the system would also create privacy risk, false positives, anxiety, overmedicalization, and opportunities for discrimination. No single payer captures the whole value, and no single actor has an uncontested right to monitor the person continuously.

Commercialization can distort the objective. If revenue depends on tests, consultations, supplements, or interventions, the system may benefit from finding more risks. If insurers operate the system, users may fear premium consequences. If employers operate it, surveillance enters the workplace. If the state operates it, public-health purpose can merge with administrative control. A fiduciary institution is needed, but the fiduciary itself needs sustainable finance.

Rare diseases expose the market-size problem. AI may reduce discovery costs by identifying targets, repurposing drugs, designing molecules, and matching patients. But clinical validation, manufacturing, and regulatory approval remain expensive. Small populations cannot always support the required investment without high prices, public subsidy, shared trial infrastructure, or alternative reward mechanisms. Orphan-drug policies attempt to correct this

through exclusivity and incentives, yet they can create tensions between innovation and affordability.

Antimicrobial resistance presents the stronger inversion already discussed: new antibiotics are socially valuable but should be conserved. Sales-based reward is misaligned with public health. Subscription payments, market-entry rewards, pooled procurement, and international pull incentives are attempts to pay for availability rather than volume. The case should be treated as a prototype for other reserve capacities. A pandemic-response platform, emergency manufacturing line, cyber-response team, or disaster stockpile is valuable because it exists before demand. Waiting for the market signal means waiting for the emergency.

Mental-health prevention is similarly fragmented. Early support, community design, reduced isolation, humane employment, and digital environments that avoid compulsive comparison could prevent distress. But health systems are often paid for treatment episodes, employers measure productivity, platforms measure engagement, and municipalities divide responsibility across housing, education, health, and policing. The cause is systemic; the budget is categorical.

Care for aging populations creates another ownerless domain. Technologies that preserve independence—home adaptation, social robots, mobility support, cognitive assistance, community coordination—can reduce institutional care and improve dignity. Savings may accrue to families, health services, long-term care insurers, and local governments at different times. The person most affected may have limited purchasing power. A venture-backed device can address a narrow transaction; the integrated care ecology remains unfunded.

Climate adaptation is the same structure at territorial scale. Flood defenses, heat-resilient housing, water systems, crop diversity, urban cooling, and emergency planning reduce future losses. Property owners, insurers, municipalities, utilities, and national governments benefit unevenly. Poor communities often face the highest risk and lowest capacity to finance adaptation. Private capital may fund assets with contracted revenue, but ecosystem restoration and community resilience produce diffuse benefits. The project needs collective authority before finance can enter.

Biodiversity preservation has even weaker ownership. A species, seed bank, soil microbiome, or ecosystem may contain future scientific and economic value, but current markets capture only fragments. Once lost, the option cannot be repurchased. Conservation finance can monetize carbon, tourism, or ecosystem services, yet measurement is incomplete and property rights may conflict with indigenous or public claims. The value is partly moral and partly unknown, making it difficult to price without pretending precision.

Infrastructure maintenance is less dramatic and more pervasive. Roads, water networks, public software, archives, research instruments, and buildings deteriorate gradually. Preventive maintenance is often cheaper than failure, but budgets favor new projects and emergency repair. Vendors may earn more from replacement than longevity. Public officials receive little credit for systems that continue functioning. AI-enabled predictive maintenance can improve diagnosis, but the institutional incentive to act on the diagnosis may remain absent.

Digital maintenance deserves particular attention. Society increasingly depends on open-source libraries, standards, public

databases, certificate systems, and security tools maintained by small teams. Their failure can affect millions of users, yet the maintainers may receive little of the downstream value. Companies rationally consume the commons and contribute unevenly. Procurement pays for applications, not the dependency graph beneath them. A public program that finances maintenance is difficult to celebrate because its success is the absence of crisis.

Long-term knowledge preservation has the same fate. Scientific data, software, model checkpoints, laboratory protocols, and institutional memory require migration, metadata, security, and stewardship. Researchers are rewarded for producing new outputs, not maintaining old ones. Grants end. Staff leave. File formats decay. A civilization capable of generating unprecedented knowledge can become incapable of reproducing it because preservation has no glamorous sponsor.

The self-erasing project extends beyond prevention. Imagine a company that automates a bureaucratic procedure so completely that its own service is no longer needed; a mediation platform that helps parties resolve conflict without repeated dependence; an educational system that makes learners autonomous; or a poverty intervention designed to eliminate its target population. These projects can be funded through outcome payments, but outcome contracts require measurable attribution and a credible payer. They also create moral hazards: providers may select easier cases, optimize the metric, or define success narrowly.

The general solution is delinkage. Payment must be separated from the volume of the harmful event or remedial transaction. Society can pay for readiness, verified risk reduction, maintained capacity, open

availability, or independently measured outcomes. Insurance can share savings. Governments can issue prevention bonds. Health systems can use capitated or value-based payments. Advance market commitments can guarantee demand for innovations meeting public specifications. Maintenance budgets can be protected as a percentage of asset value. Public data can expose deferred liabilities.

Delinkage does not remove the need for evidence. It increases it. If a provider is paid for prevention, measurement must distinguish real risk reduction from favorable selection and storytelling. AI can help integrate evidence, simulate counterfactuals, and monitor systems, but models can also manufacture false certainty. The evaluator must be independent of the provider and contestable by affected people.

The deeper institutional principle is that absence can be a product. Liberal economies are skilled at paying for things that happen. The AI age will require institutions capable of paying reliably for things that do not happen, for options that remain open, and for capacities that should rarely be used.

2. The Commons Penalty: Public Compute and Verification Infrastructure

Knowledge is unusual economic material. It is expensive to create, cheap to copy, cumulative, and often more valuable when shared. Intellectual-property systems attempt to balance incentives for creation with diffusion. Public research, universities, libraries, standards organizations, open-source communities, and scientific journals form a mixed ecology around that balance. AI destabilizes it by sharply reducing the cost of producing and recombining knowledge-like

outputs while increasing the importance of data, compute, validation, and trusted access.

A universal AI knowledge commons would include more than open model weights. It would provide machine-readable scientific literature, provenance, datasets, reusable methods, evaluation suites, model cards, safety tests, domain ontologies, executable protocols, and public compute. It would allow researchers and small organizations to build without surrendering their work to a dominant platform. Its purpose would be to make intelligence infrastructure contestable and cumulative.

No single firm naturally owns this project. A proprietary provider can offer excellent tools, but its incentives include lock-in, data capture, product differentiation, and margin. A university can host components but lacks global operational capacity. A government can finance infrastructure but may politicize access or prioritize national users. An open-source community can innovate but may lack stable maintenance. A consortium can coordinate members but risk becoming a club.

Digital public infrastructure offers a partial model. The World Bank defines DPI as foundational digital building blocks designed for public benefit, including identity, payments, and secure data exchange. OECD analysis in 2026 emphasizes that shared infrastructure can reduce duplication and support reliable services, while warning that deployment alone does not guarantee impact (World Bank, 2025; OECD, 2026a). AI infrastructure extends the concept from transactions to cognition. Public model access, evaluation, secure data collaboration, and agent interoperability may become as foundational as identity and payments.

Compute is the most visible bottleneck. Training and deploying advanced models require chips, energy, cooling, networks, and specialized engineering. Even when algorithms are open, meaningful experimentation can be limited by access. Public compute can democratize research, support strategic autonomy, and finance socially important projects that lack venture backing. But it cannot be a simple queue.

Allocation design will shape the science. Peer-reviewed access may reproduce prestige. Auctions may favor wealthy institutions. Equal quotas may waste capacity on low-value uses. Mission allocation may politicize research. Continuous AI-assisted review could reduce transaction costs, but it must avoid becoming an opaque gatekeeper. A plural system might reserve capacity for open competition, strategic missions, early-career researchers, replication, public-interest infrastructure, and rapid exploratory grants.

Data are equally important and more legally complex. Many high-value datasets involve health, education, employment, public administration, or commercial activity. Their use can improve science and services while threatening privacy and power. Data trusts, federated learning, secure computation, differential privacy, confidential computing, and controlled research environments can reduce risk. Yet these technologies often add cost and limit convenient reuse. The commercial actor with the strongest incentive to integrate data may be the actor least trusted to govern it.

A public-interest data institution would need duties to individuals, communities, researchers, and future users. It would define permitted purposes, preserve provenance, audit access, and ensure that benefits return to contributors. This is not merely a technical platform. It is a

governance system for relational assets. Its costs are continuous; its benefits escape widely.

Verification infrastructure is the next commons. As generative systems increase the volume of claims, code, images, and scientific hypotheses, society needs systems that track origin, evidence, uncertainty, and transformation. The market naturally produces content because content attracts attention. Verification is slower, adversarial, and often unwelcome to the producer. A platform may invest in trust to protect its brand, but an independent verification layer may be more credible precisely because it is not controlled by any content platform.

Scientific verification could include automated consistency checks, statistical review, code execution, replication support, literature contradiction maps, negative-results repositories, and claim-level provenance. Legal verification could trace citations, jurisdiction, amendments, and conflicts. Public-policy verification could connect claims to datasets and models. AI can perform much of the routine analysis, but the infrastructure must preserve uncertainty rather than manufacture a single authoritative answer.

This project has weak direct monetization. Researchers expect search and review tools to be inexpensive. Institutions may pay for premium services, but a proprietary model can bias access and become a new gatekeeper. Advertising is incompatible with epistemic neutrality. Subscription tiers can create unequal science. Public funding is therefore plausible, but traditional grants prefer finite projects over permanent epistemic utilities.

Cybersecurity is another ownerless layer. Individual firms invest in their own defense, but vulnerabilities propagate through shared

dependencies. Secure open-source maintenance, threat intelligence, coordinated disclosure, software bills of materials, and supply-chain verification benefit entire ecosystems. The actor who pays for finding and fixing a vulnerability may prevent losses for competitors. Liability can improve incentives, but excessive liability can suppress disclosure and open development.

Privacy infrastructure has similar spillovers. Interoperable consent, privacy-preserving analytics, identity credentials, and personal data agents can reduce systemic risk. Yet each organization may prefer its own account system and data silo. Common privacy infrastructure creates competition and user freedom while weakening proprietary advantage. It needs regulatory support, public procurement, or collective ownership to overcome the incentive to defect.

Standards are perhaps the purest example. A good standard reduces coordination cost and enables markets. Its value appears in compatible products made by others. Standard-setting can be funded by memberships, governments, or industry consortia, but dominant firms may shape standards to protect their position. AI agent protocols, provenance formats, model-evaluation interfaces, and machine-readable scientific representations will have enormous downstream effects. Leaving them entirely to firms risks enclosure; leaving them entirely to government risks slowness and national fragmentation.

Education commons belong in the same category. AI can provide personalized instruction, but high-quality curricula, verified explanations, simulations, assessments, and multilingual resources require sustained creation and review. A commercial tutor can bundle these assets, while a public commons would allow many tutors to

compete. The common layer reduces duplication and protects universal access. Its success makes the application market more competitive, which is precisely why no application provider has a natural incentive to fund it fully.

The ownership problem extends to institutional memory. Public agencies increasingly depend on vendor systems that store workflows, decisions, and data in proprietary formats. When a contract ends, the state may retain documents but lose operational knowledge. AI agents trained on organizational history could deepen this lock-in. Portable, open, auditable institutional memory should be treated as a public asset. Procurement rarely prices the future cost of dependency adequately.

The common response to these problems is “open source.” Openness is necessary in many cases but not sufficient. Someone must maintain, secure, document, govern, and support the system. The European Union’s 2026 initiative on open and interoperable digital ecosystems illustrates both the sovereignty benefits of openness and the sustainability barriers facing public adoption (European Commission, 2026a). A mature commons needs institutions, not only licenses.

Possible institutions include public-interest technology foundations with multi-year operating mandates; mandatory contributions from large commercial beneficiaries; procurement preferences for open standards; public compute credits tied to open outputs; matching funds for maintenance; data trusts; and regulated interoperability obligations. The goal is not to abolish commercial layers. It is to ensure that no firm must own the road in order to build a vehicle.

AI makes such foundations more urgent because the layer that becomes common determines who can innovate above it. If models,

compute, data, identity, and evaluation are all enclosed, apparent abundance will coexist with concentrated agency. If every layer is public, experimentation may stagnate and political control may expand. The difficult but necessary design is a mixed architecture in which the common foundations are protected, competitive complements remain open, and authority over truth is plural and contestable.

3. The Anti-Throughput Future: Technologies That Help Us Need Less

Modern innovation policy usually assumes that a successful technology creates more economic activity. It opens markets, increases productivity, enables transactions, or creates demand. A neglected class of technologies creates value by helping people need less: less energy, less replacement, less persuasion, less bureaucracy, less intermediation, less work, less medical treatment, less time on platforms, and less dependence on experts or vendors.

These technologies are economically subversive because gross domestic product and corporate revenue measure activity more readily than avoided activity. A durable product may create more lifetime value than several disposable products while generating fewer sales. An AI agent that resolves a legal issue in minutes may improve access to justice while reducing billable hours. A scientific assistant that prevents redundant experiments may reduce grant expenditure. A personal purchasing agent that recommends “buy nothing” may improve household welfare while shrinking commerce.

The anti-consumption agent is a useful thought experiment. It has access to the user's goals, finances, existing possessions, repair options,

health, and long-term preferences. When shown an advertisement, it calculates total cost, likely use, durability, opportunity cost, environmental impact, and whether the purchase is an attempted response to boredom or status anxiety. It negotiates, compares, recommends used goods, or advises waiting.

Such an agent is valuable to the user and hostile to much of the advertising economy. The platforms best positioned to distribute it may earn from transactions or attention. Retailers will not sponsor an agent that systematically reduces sales. A subscription model could work for affluent users, but universal deployment would require a public or cooperative option. Financial institutions might fund it if reduced spending improves repayment, yet that creates conflicts when they also sell credit and investment products.

Repair technologies have a similar structure. AI diagnostics, modular hardware, open manuals, local fabrication, and parts marketplaces could extend product life. They reduce waste and household cost. Manufacturers may profit from services, but durable repairable products can cannibalize replacement revenue. Right-to-repair laws change incentives, yet legal access to parts and information does not automatically finance the repair ecosystem or ensure safety.

Energy-efficiency systems save money and emissions, but split incentives persist. A landlord pays for insulation while a tenant pays energy bills. A factory manager may reject an upgrade with a longer payback than internal capital rules permit. A public building may have separate operating and capital budgets. AI can identify savings precisely without creating the authority to capture them.

Administrative simplification is another anti-rent project. Citizens and firms spend time navigating forms, eligibility rules, permits, taxes, and compliance. AI could translate legal requirements, pre-fill applications, detect redundant requests, and redesign services around life events. The social gain is large, but parts of the professional ecosystem earn from complexity. Agencies may also resist simplification because forms encode controls and organizational boundaries. A private “bureaucracy assistant” can charge users to navigate the maze, creating a business dependent on the maze’s persistence.

The stronger project is to eliminate the maze. That requires government to share data safely, harmonize rules, and accept responsibility for proactive service. The vendor that helps eliminate the process may lose future revenue. Outcome-based procurement could pay for time saved and procedures retired, but agencies must measure the baseline and protect rights of appeal. Simplicity can become coercive if the system removes human discretion or assumes a single normal life path.

Legal and scientific intermediaries face related changes. An AI system that makes contracts, regulations, and evidence comprehensible can democratize expertise. It can also reduce the rents created by opacity. Professionals provide genuine judgment and responsibility, so complete disintermediation is neither possible nor desirable. The beneficial technology separates routine translation from accountable counsel. Funding it threatens business models built on bundling the two.

A fiduciary AI offers a general solution. The agent would be legally obligated to place the user’s interests ahead of the provider’s, disclose conflicts, preserve confidentiality, and support portability. It could

mediate shopping, health, education, finance, employment, and digital life. The model is analogous to professional duties, but more demanding because the agent acts continuously across domains.

The economic conflict is immediate. A provider earns more if the agent recommends affiliated products, retains the user, collects data, and discourages switching. A true fiduciary must sometimes advise the user to leave. The obligation reduces monetization options and therefore valuation. The service may require direct payment, public subsidy, cooperative ownership, or regulated fees. People with the greatest need may be least able to pay, creating a new inequality in cognitive representation.

Humane social media is another difficult project. A platform could hide popularity metrics, limit algorithmic amplification, provide chronological or user-controlled feeds, slow viral spread, disclose targeting, and optimize for meaningful connection rather than time spent. Many design elements are technically straightforward. The challenge is that network effects reward scale, advertising rewards attention, and users are attracted to the very status signals they later report disliking.

A public-service social network might be financed like broadcasting or infrastructure. It would face accusations of political bias, censorship, inefficiency, and unfair competition. A cooperative network would struggle to acquire users and moderate globally. A subscription network risks becoming an elite enclave. Interoperable protocols can reduce lock-in, but moderation and discovery remain costly. The unbuildable quality comes from the combination of economic and legitimacy constraints, not lack of software.

Scientific and educational technologies can also reduce status rents. Competency-based certification could allow people to demonstrate ability without purchasing a long credential. AI tutors could personalize learning at low cost. Automated research tools could enable small teams to produce work once requiring large institutions. These are socially attractive and institutionally disruptive because universities and professions allocate more than knowledge. They distribute prestige, access, and trusted certification. A better learning tool does not automatically create an accepted signal.

Companies that deliberately refuse growth belong to the same family. A firm might decide that after reaching a sufficient scale, productivity gains will reduce prices, working hours, or environmental impact rather than increase expansion. In isolation, this is legal. In competition, it is fragile. Rivals can reinvest, acquire distribution, poach talent, and purchase the restrained firm. A mission lock helps, but the ecosystem still selects for expansion.

A company designed to dissolve after solving a problem is more radical. It would define measurable completion conditions, transfer remaining assets to a commons, and terminate. Such an organization has no conventional terminal value. It could be funded by prizes, outcome contracts, or donors, but not ordinary equity. It is an institutional form for missions rather than markets.

The common feature of these projects is not anti-capitalism. It is anti-throughput. They seek welfare without maximizing the number of transactions passing through the organization. A mature economy may need more such institutions as AI lowers production costs and material throughput becomes environmentally constrained. Yet financial

systems, corporate valuation, advertising, employment, and tax revenues remain tied to activity.

The transition will require new metrics and payment channels. Time saved, risk avoided, durability, autonomy, portability, and reduced manipulation must become contractible outcomes. Public procurement can pay for retired procedures. Manufacturers can be rewarded for product life. Platforms can be regulated around fiduciary duties. Social insurance can share prevention savings. Tax systems can shift away from labor and productive transactions toward rents, externalities, or scarce resources.

None of these mechanisms abolishes human desire for novelty or status. People may continue to buy, compare, and compete even when a rational agent advises otherwise. The point is not to optimize citizens into asceticism. It is to make a non-extractive option institutionally available. Today the system can easily finance technologies that intensify desire. It has far fewer natural pathways for technologies that help desire become optional.

CHAPTER FOUR

The Veto Society

A feasible project can survive finance and still fail in the architecture of liberal democracy. Rights, veto points, liability, organized interests, and strategic rivalry slow arbitrary power and protect plural lives. They also make certain transitions nearly impossible even when aggregate benefits appear large.

1. Veto Asymmetry: Diffuse Benefits and Concentrated Losses

Liberal democracy is designed to make power contestable. Laws pass through multiple institutions. Courts protect rights. Agencies follow procedure. Regions, professions, firms, civil society, and citizens possess formal or informal vetoes. This slows arbitrary government and allows plural interests to coexist. It also makes some beneficial transitions difficult when gains are dispersed and losses concentrated.

A reform that gives ten million people a small future benefit may face a sector that expects immediate existential loss. The beneficiaries have little reason to organize; the losers organize intensely. This asymmetry is not proof of corruption. It is a basic property of collective action. A person who expects to save a few hours per year from administrative simplification will not lobby with the intensity of a profession whose revenue depends on the procedure.

Technological change repeatedly creates this structure. Automated legal assistance can reduce routine fees while threatening established

practices. Competency-based credentials can lower education costs while weakening institutional brands. Public compute can broaden access while competing with private providers. Right-to-repair rules benefit consumers diffusely and impose concentrated adaptation costs on manufacturers. Open standards reduce lock-in while threatening platform rents. AI-assisted grant evaluation can save researcher time while reducing the influence of reviewers, consultants, and experienced grant-writing organizations.

The democratic process is often judged against an imaginary social planner who knows the optimal outcome and implements it. This comparison is unfair. The planner does not exist, and concentrated power can cause catastrophic harm. Veto points are a feature when the proposed optimization is wrong, coercive, or irreversible. The problem is that institutions cannot know in advance whether resistance protects liberty or protects rent.

Incumbents possess several advantages. They have organized associations, technical expertise, established relationships with regulators, credible claims about employment and stability, and the capacity to frame change as risk. New beneficiaries are often hypothetical. A proposed public-interest AI infrastructure promises future competition; the incumbent vendor provides a functioning system today. A new drug-payment model promises long-term innovation; current procurement rules are legally tested. A reform to peer review promises better selection; the old process has recognized experts and appeal procedures.

Regulatory expertise can become a form of political capital. Complex rules may be necessary for safety, but compliance costs favor large organizations. A startup can invent a medical or financial

technology and still lack the legal, clinical, security, and procurement infrastructure needed to deploy it. Investors anticipate this cost and avoid projects whose path to legality is uncertain. The law does not forbid the innovation; it places it behind a fixed-cost wall.

This is particularly severe for technologies that cross categories. A personal AI health agent may be simultaneously software, medical device, data controller, insurer interface, consumer product, and professional service. Each category imposes obligations designed for a world in which those functions were separate. A small firm must either choose a narrow claim, partner with incumbents, or bear regulatory complexity that overwhelms product development.

Public agencies face their own incentives. Officials receive asymmetric feedback. A visible failure can produce scandal, audit, or career damage. A missed opportunity is counterfactual and rarely attributable. This encourages safe procurement, incremental regulation, and pilot programs. Democratic accountability, intended to protect the public, can become a selection pressure against experiments whose failure would be informative.

Election cycles amplify the bias. Climate adaptation, preventive health, educational reform, scientific infrastructure, and long-term AI safety may produce benefits beyond the current government. Construction, subsidies, and emergency response produce immediate credit. Even independent agencies operate within political budgets. Future citizens are affected but cannot vote; non-human ecosystems have no direct representation; global beneficiaries are outside the constituency.

The result is not simple short-termism. Governments do make long-term investments. Pensions, infrastructure, defense, universities,

and public health demonstrate institutional patience. The difficulty arises when long-term projects require present groups to surrender rents, authority, or identity. A new bridge can distribute contracts. A new institutional layer may eliminate them.

Automation and abundance make the conflict more severe. Suppose AI dramatically increases productivity. Society could convert the gain into higher profits, lower prices, shorter working time, public services, or dividends. Each distribution creates different winners. Firms that own models and automation may prefer capital returns. Workers may prefer wages and security. consumers may prefer lower prices. Governments may prefer tax revenue and employment. A universal dividend requires a claim on the productive assets; shorter work requires coordination or bargaining; public services require taxation. Technology does not select among these outcomes.

The same applies to essential goods. If automated production makes basic education, diagnostics, legal guidance, or energy much cheaper, society could move some services outside ordinary markets. Incumbent providers may argue, often plausibly, that universal public provision will reduce quality, choice, and innovation. Advocates may argue that market access is no longer necessary. The conflict is not between abundance and irrational scarcity. It is over who governs the abundant system and who loses the right to charge for access.

Land and location reveal the limit of material abundance. AI can design buildings, optimize transport, and automate construction, but it cannot create more central land. Reforms that allow housing construction produce broad affordability benefits and concentrated neighborhood costs. Existing owners are organized, while future

residents are absent. Liberal property and local participation rights can preserve exclusion even when construction technology improves.

Inheritance presents a similar contradiction. A democracy may value equality of opportunity while protecting the family's right to transmit wealth. Each parent has a strong motive to help children; collectively, inheritance preserves positional advantage. Radical limits could improve starting equality but violate deeply held understandings of family and ownership. No technical abundance removes the conflict because the inherited good is often relative position.

Institutional reform must therefore account for transition, not only destination. A proposal that eliminates rents without compensating or repurposing the people and organizations built around them will face rational resistance. Compensation can be morally controversial—why pay actors to surrender an unjust privilege?—but politically necessary. Training, phased conversion, buyouts, public equity, grandfathering, and shared upside can turn veto players into participants.

Yet compensation has limits. If every incumbent must be made whole, transformative reform becomes unaffordable and inequality is frozen. Some losses are the normal consequence of competition. The design question is which interests deserve protection: basic security, legitimate reliance, and earned investment may; indefinite monopoly, avoidable opacity, and status privilege may not. Liberal democracy has no formula for this boundary. It negotiates it through politics.

AI could improve negotiation by modeling distributional effects, identifying hidden beneficiaries, and designing transition options. It could also strengthen incumbents by producing more persuasive lobbying, regulatory comments, and targeted campaigns. Scalable

intelligence does not dissolve organized power. It can make power more computational.

The institutionally impossible project often fails here, after technical feasibility and before formal prohibition. Everyone agrees on aggregate benefit; no coalition accepts the transition. The future has a social surplus but no political owner.

2. Law, Liability, and the Right to Refuse Optimization

Law is frequently portrayed as a barrier to innovation. Innovators complain that categories are outdated, approvals slow, and liability uncertain. These complaints can be valid. But law performs a function that technical systems do not: it decides whose interests count when optimization causes harm. It assigns responsibility, protects rights, defines acceptable evidence, and gives people a language for refusal.

The AI age creates systems that are not merely tools but decision environments. A medical agent influences treatment. An educational agent shapes a child's development. A hiring model affects opportunity. A personal assistant filters information and purchases. An autonomous vehicle allocates risk. A scientific reviewer influences careers and knowledge. Each system can be statistically superior to ordinary human practice while remaining unacceptable in particular cases.

Suppose autonomous vehicles become ten times safer than human drivers. Is manual driving then a private liberty or a public hazard? If an AI diagnostic system reduces average mortality but occasionally fails in novel ways, should hospitals be required to use it? If personalized education produces better outcomes but requires continuous cognitive

monitoring, can parents refuse? If an automated grant system is demonstrably more consistent than panels, does a researcher have a right to human review?

These are not performance questions. They concern the moral status of error. People often tolerate human imperfection differently from machine imperfection because human decisions contain recognizable intentions, responsibility, and the possibility of mercy. A human judge can explain, apologize, or be sanctioned. An automated system distributes causality across developers, data, deployers, and users. Superior average performance does not answer who bears the exceptional harm.

Liability uncertainty can make valuable technologies economically infeasible. If every actor may be responsible, insurers price extreme risk or refuse coverage. If no actor is responsible, victims lack remedy and trust collapses. Product liability, professional liability, data protection, contract, and administrative law may each apply partially. The innovation sits in a gap where rights are real but roles undefined.

Large incumbents can survive this uncertainty through legal departments, reserves, and influence. Small entrants cannot. Regulation therefore has a paradoxical effect. Strong rules can create trust and markets, but complex compliance can entrench the firms most capable of bearing it. A path to legality that costs tens of millions is formally open and economically closed.

Regulatory sandboxes attempt to create controlled experimentation. They allow temporary waivers, close supervision, and learning before general rules. Sandboxes are useful for discovering risks and building regulator capacity. They do not automatically solve the transition to scale. A pilot with selected users and special oversight may

succeed under conditions impossible to universalize. Regulators still need a general liability regime, interoperability rules, audit standards, and enforcement capacity.

Adaptive systems create a further problem. Traditional approval evaluates a product version. A learning system changes through data, updates, model replacement, and interaction. Freezing it sacrifices improvement; allowing continuous change weakens the meaning of approval. Regulation must move from artifact certification to process assurance: update controls, monitoring, rollback, provenance, testing, and post-deployment evidence. This is more technically demanding and more dependent on access to systems that firms may consider proprietary.

Trade secrecy collides with due process. A company may protect model details, data, and evaluation methods as competitive assets. A person denied credit, employment, benefits, or treatment needs enough explanation to contest the decision. An institution cannot outsource public authority to a system whose reasons are commercially inaccessible. Yet full disclosure can expose security vulnerabilities, privacy, and intellectual property. The solution requires trusted auditors, standardized evidence, and rights to contest outcomes without demanding publication of every implementation detail.

Optimization also threatens pluralism. A liberal society does not agree on one definition of flourishing. A health system may optimize longevity, quality of life, autonomy, equality, or cost. An education system may optimize test performance, curiosity, civic competence, or emotional development. A social platform may optimize informed participation, connection, free expression, or psychological well-being.

AI makes multi-objective trade-offs explicit but cannot legitimately settle them.

The danger of a highly capable state is not only surveillance. It is benevolent overreach. A government able to predict harmful choices may nudge, restrict, or preempt them. Insurance can become behavior management. Public health can become continuous monitoring. Education can become personality optimization. Carbon policy can become individualized consumption control. Each intervention can be justified by aggregate evidence.

Liberalism therefore needs a right to refuse optimization. This does not mean a right to impose unlimited externalities. A person cannot refuse traffic rules while endangering others. But where harms are primarily personal or values genuinely plural, citizens may need rights to human review, local processing, non-personalized service, data minimization, explanation, exit, and deliberate inefficiency. The right to make a suboptimal choice may become central to autonomy.

There may also be a right not to know. Predictive medicine can reveal future disease risk. Genetic analysis can expose family information. Behavioral models can infer addiction, political vulnerability, or relationship instability. More information is not always beneficial, and individuals may prefer uncertainty. A system optimized for prevention can violate that preference unless consent is meaningful.

Consent itself becomes fragile under pervasive AI. Terms are complex, services essential, and predictions difficult to understand. Clicking agreement cannot legitimize every use. Fiduciary duties, purpose limitation, collective governance, and default protections are stronger than transactional consent when power is asymmetric.

The law will also confront artificial persons or person-like systems. If future AI systems exhibit persistent preferences, self-models, or credible signs of experience, treating them solely as property may become morally unstable. Granting rights would transform economic models built on ownership and replication. Even before consciousness is established, society may adopt precautionary limits against creating systems that could undergo vast amounts of suffering. Such limits would be difficult to verify and strategically costly if rivals ignore them.

Biotechnology raises parallel conflicts. Embryo selection, gene editing, artificial gestation, and neural enhancement may reduce disease or expand choice while challenging inherited categories of parenthood, equality, and human identity. A liberal framework based on individual consent struggles when choices affect future persons who cannot consent and alter competitive conditions for everyone. If enhancement is allowed privately, inequality may become biological. If prohibited, preventable suffering may continue. If publicly provided, the state enters the design of future citizens.

Law cannot make these contradictions disappear. Its role is to convert them into rights, duties, procedures, and limits that people can contest. This necessarily slows deployment. The unbuildable future sometimes deserves to remain unbuilt until those protections exist. Institutional impossibility is not always a defect; it can be a warning that technical capability has outrun legitimate authority.

The challenge is to distinguish protective friction from inherited obstruction. A good legal pathway makes the cost of safety proportionate, permits staged evidence, assigns liability clearly, preserves appeal, and scales requirements with risk. A bad pathway is formally neutral but so opaque and expensive that only incumbents can

use it. AI can help draft, simulate, monitor, and enforce better rules. It cannot decide which freedoms are worth the inefficiency they create.

3. The Restraint Race: Why Rational Actors Cannot Stop Together

Some projects fail not because no actor wants them, but because no actor can pursue them alone without transferring advantage to others. Climate mitigation, AI safety, cybersecurity, biosecurity, tax coordination, autonomous weapons, privacy, and open science all contain versions of this dilemma. Cooperation would improve collective welfare; unilateral cooperation can be strategically irrational.

The logic is familiar. A state may prefer that no one build autonomous weapons, but if rivals build them, abstention can create vulnerability. A company may prefer rigorous AI safety standards, but if competitors deploy faster under weaker rules, restraint can destroy market position. A country may tax mobile capital more heavily, but capital can relocate. A researcher may disclose safety findings openly, but adversaries may exploit them. A platform may limit persuasive targeting, but users and advertisers may migrate to a less restrained rival.

These are not ordinary market failures because the relevant market includes security. States value relative capability, not only absolute welfare. A technology that makes everyone richer can still be destabilizing if it shifts power. AI is especially sensitive because it may affect research, military planning, cyber operations, surveillance, propaganda, and industrial productivity simultaneously.

Scalable intelligence can accelerate races in two directions. It lowers the cost of developing and adapting capabilities, shortening the time

available for governance. It also improves monitoring, verification, and coordination, potentially making agreements more enforceable. The same technology that enables autonomous cyberattacks can audit networks; the same models that design biological agents can detect them; the same surveillance that verifies treaty compliance can threaten civil liberty.

Open-source AI illustrates the conflict. Openness supports scientific scrutiny, competition, education, and distributed innovation. It reduces dependence on a few providers. It can also lower barriers for malicious use and make capability controls difficult. Closed systems concentrate power and obscure risks. Neither openness nor closure is universally safe. The correct policy may differ by capability, deployment context, and availability of defensive tools.

Cybersecurity has long lived with this dual-use structure. Publishing vulnerabilities enables repair and enables attack. Hoarding vulnerabilities can support intelligence operations while leaving citizens and firms exposed. Defensive investment creates positive externalities, but offensive capability may be strategically prized. International norms exist, yet attribution and enforcement remain difficult.

Biosecurity can be more severe because harms may cross borders irreversibly. Screening DNA orders, monitoring laboratories, securing pathogen data, and coordinating outbreak response require shared standards. States may resist transparency about facilities and incidents. Firms may treat sequence data and detection methods as proprietary. Low-capacity countries may view requirements as barriers imposed by rich countries. A global safety regime needs resources and legitimacy, not only rules.

Climate policy demonstrates the distribution problem. Every state benefits from mitigation, but each bears domestic costs and has incentives to free-ride. Trade measures, carbon clubs, technology transfer, and climate finance attempt to align incentives. AI may improve energy systems and climate modeling, but it can also increase electricity demand and accelerate consumption. The technical solution does not create the political bargain.

Global tax and data governance reveal a similar problem. Digital firms operate across jurisdictions, while taxation, privacy, and competition law remain largely national or regional. A country that imposes strict rules may protect citizens but reduce local investment. A country that offers permissive conditions can attract firms while exporting risks. Regulatory competition can produce a race to the bottom or a fragmented world of incompatible regimes.

Liberal democracies face an additional constraint. International coordination can weaken domestic accountability. Treaties and technical standards are negotiated by executives, diplomats, and experts. Citizens affected by global rules have limited ability to remove the decision-makers. Strong global institutions can solve coordination problems while creating legitimacy deficits. Weak institutions preserve sovereignty while failing to control transnational technology.

The common proposal is an international agency for AI or other frontier risks. Such an agency could set standards, share incident data, coordinate evaluation, and inspect high-risk facilities. It would immediately face questions of representation, secrecy, enforcement, and unequal capability. States with advanced technology would resist transferring strategic information. Less powerful states would fear

permanent exclusion. Firms would lobby over standards. Authoritarian and democratic governments would disagree on rights.

Coordination can be built in narrower layers. Clubs of willing states can harmonize evaluation and procurement. Mutual recognition can reward compliant firms. Shared compute or research facilities can create benefits for participation. Insurance and finance can require safety standards. Export controls can slow diffusion of particular capabilities, though they also encourage substitution and fragmentation. Advance commitments can finance defensive technologies as global public goods.

Verification is central. Agreements survive when compliance can be observed and violations have consequences. AI may improve anomaly detection, model auditing, satellite analysis, and scientific monitoring. But verification systems themselves must be trusted and protected from manipulation. A dominant state's proprietary model cannot easily serve as neutral global inspector.

The race problem also appears inside science. Researchers and organizations may recognize that rapid publication or deployment is risky but fear being preempted. Priority, citations, patents, and funding reward being first. Safety work and replication slow the race without equivalent prestige. A coordination regime must change rewards, perhaps through shared credit, precompetitive consortia, confidential disclosure channels, and funding for verification.

The deeper impossibility is that restraint is a public good while capability is an asset. Actors can own capability and only collectively own restraint. This makes restraint fragile. It must be supported by institutions that convert collective safety into local advantage: market

access, insurance, procurement eligibility, shared research, reputational benefit, or reciprocal security.

There are limits to incentive design. Some actors value dominance more than collective gain. Some cannot be verified. Some technologies are too distributed to control. The goal is not perfect coordination but reduction of the most dangerous races and creation of defensive abundance. Public investment in detection, resilience, medical countermeasures, secure infrastructure, and rapid recovery can reduce the payoff to attack even when offensive control fails.

The global unbuildable future is therefore double. Humanity struggles to build shared protections, and it struggles to refrain from building dangerous capabilities. Both failures arise because “humanity” is not a sovereign actor. The institutions capable of acting have narrower constituencies and competitive obligations.

Any serious book about beneficial technology must therefore resist a comforting assumption: that once a superior design is demonstrated, rational actors will converge on it. In strategic environments, the superior collective design may be an inferior unilateral move. The missing technology is often credible reciprocity.

CHAPTER FIVE

Abundance Meets the Human Animal

AI is often imagined as a machine for dissolving scarcity. Even if it helps produce material abundance, it cannot multiply every good that humans value. Status, exclusive position, admiration, trusted identity, intimate attachment, and legitimacy remain relational. The social order can therefore resist abundance for reasons deeper than finance.

1. The Positional Ceiling: What Abundance Cannot Multiply

The strongest promise of artificial intelligence is not that it will imitate individual human tasks. It is that some forms of useful cognition may become reproducible at very low marginal cost. Combined with robotics, biotechnology, advanced materials, and abundant energy, scalable intelligence could reduce the cost of designing, coordinating, and producing a wide range of goods. Education, legal guidance, software, diagnosis, entertainment, scientific analysis, and administrative work may become substantially cheaper. In optimistic scenarios, material scarcity ceases to organize everyday life as completely as it does today.

This would be a civilizational achievement. It would not, however, abolish competition. Human beings do not compete only for calories, shelter, or information. They compete for rank, attention, authority,

admiration, desirable partners, proximity to influential people, and membership in groups that confer distinction. These are positional goods: their value depends partly on who else has them and where the holder stands relative to others. A society can manufacture more comfortable homes, but it cannot give everyone the most prestigious address. It can offer excellent education to all, but it cannot make everyone more credentialed than everyone else. It can expand the supply of entertainment, but not the supply of being widely watched.

The distinction between material and positional scarcity matters because many technological utopias quietly assume that reducing the first will dissolve the second. The assumption is psychologically implausible and institutionally dangerous. If basic goods become abundant while status remains scarce, competition may migrate rather than disappear. People may spend less effort securing subsistence and more effort securing differentiation. Luxury can move from objects to experiences, identity, authenticity, privacy, biological enhancement, exclusive communities, or access to human attention. The object changes; the ranking function survives.

AI may intensify this process. Generative systems lower the cost of producing competent text, images, music, software, analysis, and design. As average quality rises, the scarce resource becomes selection and recognition. Millions of people can produce polished work; very few can command a large audience. Creation becomes abundant while trusted attention becomes more concentrated. The result may be a paradoxical economy in which more people can express themselves and fewer people feel seen.

The same effect can appear in professional life. When AI makes routine competence widely available, distinction migrates toward

difficult-to-measure qualities: judgment, taste, leadership, reputation, access, originality, courage, or perceived authenticity. Some of these are genuine capabilities. Others are socially constructed filters that preserve hierarchy when older filters lose power. Credential inflation is one example. When more people obtain a qualification, institutions demand a higher one. When AI makes standard analytical work cheap, employers may value prestigious affiliations more heavily because they remain scarce signals.

This does not mean status competition is wholly pathological. Rank can organize responsibility, reward contribution, and motivate effort. Recognition can signal trust. Ambition can generate discovery and art. The problem arises when institutions confuse a useful allocation signal with a measure of human worth, or when positional competition consumes the gains from material abundance. A society may become richer while its citizens remain trapped in relative deprivation.

Housing illustrates the distinction. Better construction and automation could reduce building costs dramatically. Yet land in desirable locations remains finite, and desirability itself is socially produced. A neighborhood becomes valuable because of schools, safety, amenities, beauty, networks, and the people already admitted. Increasing the supply of housing can reduce material deprivation, but exclusive zoning and status sorting can preserve positional scarcity. The technology to build is not the institution that grants access.

Education presents a similar case. Universal AI tutoring could provide instruction of extraordinary quality. This would be a major public benefit. But elite education also functions as selection, network formation, and status certification. If everyone receives excellent

instruction, the scarcity may move to selective cohorts, internships, recommendations, competitions, and proximity to prestigious mentors. The declared function of education can become abundant while its hidden function remains scarce.

Sexual and romantic competition adds a still more sensitive dimension. Material security can improve relationships and reduce coercion, but it cannot guarantee equal desirability or partnership outcomes. Digital platforms already expand the apparent field of comparison and concentrate attention on a minority of highly visible profiles. AI-generated companions, matching systems, cosmetic enhancement, and reputation analytics could reduce loneliness for some while producing new hierarchies for others. A society that promises abundance without acknowledging unequal romantic outcomes may discover that biological and emotional competition has become a central political grievance.

The phrase biological competition should be used carefully. Human behavior is not a fixed script dictated by genes. Culture, law, institutions, and personal commitments profoundly alter how status and mating competition are expressed. Yet it is equally mistaken to assume that competition is merely an ideological artifact that disappears when economic arrangements change. Humans compare themselves, form coalitions, seek recognition, protect kin, and respond strongly to perceived humiliation. Institutional design must work with these tendencies without treating them as destiny.

A serious abundance policy therefore requires two layers. The first guarantees material sufficiency: health, shelter, energy, education, mobility, communication, and access to intelligence. The second limits the conversion of positional advantage into domination over basic life

chances. Progressive taxation, inheritance rules, antitrust, public education, universal services, labor protections, and democratic rights already perform this second function imperfectly. In an AI economy, new versions may be needed for compute, data, visibility, enhancement, and access to scarce human institutions.

The goal cannot be equal status, which is logically impossible if status is relational. It can be plural status: many domains of recognition, multiple paths to esteem, and fewer winner-take-all rankings. A society is more stable when excellence in one domain does not purchase total superiority in all others. Scientific authority should not automatically buy political power; wealth should not buy immunity from law; online popularity should not determine educational opportunity; biological advantage should not purchase unlimited reproductive or cognitive enhancement for descendants.

Platforms can help or harm this pluralism. A single global ranking system collapses distinct values into visibility. Recommendation algorithms often amplify already visible actors because popularity predicts engagement. Alternative designs can diversify exposure, limit cumulative advantage, create local communities, and separate recommendation from public status counts. These interventions will not abolish comparison, but they can prevent one metric from becoming the universal currency of worth.

The political danger is not simply inequality. It is the belief that abundance was promised and personally withheld. When material production becomes visibly easier, continued exclusion appears less like fate and more like deliberate allocation. People may tolerate scarcity caused by nature more readily than scarcity produced by ownership rules, zoning, artificial limitations, or status closure. AI can therefore

increase legitimacy pressure even before it produces abundance: it makes alternative arrangements easier to imagine.

This creates a category of unbuildable project that conventional technology policy rarely recognizes. The project is not a new product. It is an institutional arrangement that captures productivity gains without allowing all value to be absorbed by positional competition. Examples include public AI dividends, universal access to high-quality tutoring, noncommercial recommendation infrastructures, public-interest identity systems, limits on attention markets, and inheritance mechanisms that prevent permanent caste formation. Each is technically modest compared with frontier AI. Each is politically difficult because it redistributes not only money but rank.

A post-scarcity society, if that phrase is ever deserved, will still contain scarcity. The important question is which scarcities remain, who is permitted to manufacture them, and whether they govern access to a dignified life. Technology can multiply goods. It cannot multiply first place. The future becomes humane only when institutions stop treating every positional contest as the natural method for distributing material security.

2. Work, Merit, Family, and Inherited Advantage

The public debate about AI and work is usually framed as a problem of employment. Will jobs disappear, change, or multiply? Which skills will remain valuable? How quickly can workers retrain? These are important questions, but they are narrower than the institutional problem. Work in liberal societies is not only a method of

production. It is a distribution mechanism, a moral category, a source of identity, a daily structure, a community, and a claim to social membership. Automation can reduce the need for labor without automatically replacing any of these functions.

This is why a society may preserve unnecessary work even when technology could eliminate it. Income is attached to employment. Health insurance, residency rights, pensions, creditworthiness, and social respect may also be attached to employment. Managers derive authority from directing labor. Governments depend on payroll and income taxes. Cities organize transport and land use around workplaces. Families structure time around school and work schedules. A reduction in necessary labor threatens an entire institutional ecology.

The familiar prediction that higher productivity will produce more leisure therefore has no automatic mechanism. A firm that becomes twice as productive does not normally halve working time while preserving pay. It may expand output, reduce headcount, raise margins, lower prices, acquire competitors, or increase expectations. Competition converts productivity into advantage before it converts it into leisure. Without collective bargaining, working-time regulation, taxation, public ownership, or new distribution rules, the gain is allocated by property and bargaining power.

AI makes this problem sharper because it can automate components of cognitive work that have long justified professional status. A person may remain employed while much of the productive activity is performed by systems owned by the firm. The employee becomes an interface, supervisor, accountable signatory, relationship manager, or institutional bearer of liability. Compensation may remain temporarily high because responsibility is scarce, then fall as processes

standardize. The economic value of labor and the social meaning of occupation can separate.

A humane response cannot consist only of retraining everyone for the remaining scarce jobs. That strategy assumes the objective is to keep all adults competing for employment regardless of whether their labor is socially necessary. It also converts every productivity gain into a demand that people reinvent themselves faster. Education becomes permanent defensive adaptation to machines rather than development of human capability.

The more radical project is the deliberate conversion of productivity into discretionary time. This could take the form of shorter standard workweeks, job sharing, sabbatical rights, universal basic services, social dividends, participation income, or public ownership of automated capital. Each mechanism has advantages and risks. A cash-only basic income may subsidize rents if housing remains scarce. Universal services may become paternalistic if choice is weak. Public ownership can be inefficient or politically captured. Shorter hours can intensify work if output expectations remain unchanged. The point is not that one instrument solves the problem. It is that leisure must be institutionally allocated; technology does not distribute it by itself.

A related difficulty is merit. Liberal societies justify inequality partly through the idea that rewards correspond to effort, talent, contribution, or responsible choice. The principle is never fully realized, but it legitimizes outcomes. AI destabilizes it. If output depends increasingly on access to models, compute, data, capital, and organizational position, individual contribution becomes difficult to isolate. A founder may create enormous value by coordinating AI

systems while employing few people. A worker may perform excellent work through tools unavailable to others. A student may learn rapidly with a personalized tutor. Are these achievements merited, inherited, rented, or collectively produced?

The question is not merely philosophical. Taxation, education, welfare, intellectual property, and employment law depend on implicit theories of contribution. When scalable intelligence is trained on collective knowledge and deployed through privately owned infrastructure, the boundary between individual accomplishment and inherited civilizational capital becomes unstable. A social dividend can be defended not as charity but as a return on shared intellectual inheritance.

Family intensifies the conflict between equal opportunity and legitimate partiality. Parents reasonably wish to help their children. They transmit money, housing, education, networks, language, confidence, health, and cultural knowledge. Liberal equality accepts much of this as part of family freedom while attempting to prevent inherited privilege from becoming absolute. The compromise is already unstable. AI tutoring, genetic selection, neurotechnology, longevity, and private access to advanced healthcare could magnify inherited advantage beyond the capacity of ordinary education and taxation to compensate.

Suppose safe cognitive enhancement becomes available. Allowing parents to purchase it can create biological class stratification. Prohibiting it may deny children health or capability benefits. Public provision may reduce inequality while involving the state in intimate decisions about desirable traits. The same trilemma appears in embryo selection, longevity treatments, reproductive technology, and

continuous developmental monitoring. Market freedom, equality, and plural moral autonomy cannot all be maximized simultaneously.

Inheritance is the conventional version of the problem. Strong inheritance rights express family continuity and property freedom. Strong estate taxation protects mobility and prevents dynastic power. Both positions appeal to legitimate values. AI-generated wealth could increase the stakes if a small number of owners capture returns from automated capital for generations. A society with universal material abundance could still become hereditary in status, influence, land, and enhancement.

The unbuildable project here is genuine equality of opportunity. It is widely endorsed and structurally resisted. Achieving it would require limiting precisely the activities through which families express care and successful actors secure continuity. It would also require compensating for unequal biology, location, health, and early childhood environments at a depth that many citizens would experience as intrusive. No liberal democracy can completely neutralize family advantage without ceasing to be liberal, and no society can tolerate unlimited inheritance without risking oligarchy.

A more defensible objective is to establish a high social floor and a ceiling on the conversion of inherited advantage into domination. Universal access to excellent healthcare, tutoring, compute, cultural participation, and safe environments can reduce brute inequality. Progressive taxation and competition policy can prevent compounding ownership from becoming sovereignty. Public endowments for every child can socialize part of capital inheritance without abolishing private families. Rules can distinguish therapeutic enhancement from status enhancement, although the boundary will remain contested.

The right not to compete may become as important as equality within competition. Today, opting out of labor-market competition often means poverty, dependence, or loss of status. In an abundant economy, a person should be able to choose care, study, art, community work, contemplation, or a modest private life without forfeiting basic security. This is not a right to admiration or luxury. It is a right to dignified nonparticipation in races whose economic necessity has diminished.

Such a right conflicts with deeply rooted moral intuitions. People who work may resent supporting those who do not. Citizens may fear free riding. Political communities distinguish the deserving from the undeserving. These intuitions are partly responses to real scarcity and partly mechanisms of status. AI abundance may weaken their material basis long before it weakens their emotional force.

The transition therefore requires a new language of contribution. Care, learning, maintaining relationships, preserving ecosystems, participating in democracy, producing open knowledge, and supporting community resilience are socially valuable even when they do not generate market wages. At the same time, declaring every activity a contribution can empty the concept. Institutions will need plural recognition without recreating a totalizing social-credit system.

The worst response would be to fabricate employment solely to preserve the moral order of work. Meaningless compliance tasks, performative meetings, artificial credential barriers, and bureaucratic supervision can absorb human time without producing corresponding value. AI may make such work more visible, but it can also automate production while expanding oversight and documentation. The

bureaucracy of proving that one is productive can survive the decline of production itself.

A better future treats work as one valuable mode of participation rather than the universal license to exist. It distributes part of automated productivity as time, capability, and security. It accepts family attachment while preventing hereditary sovereignty. It rewards contribution without pretending that every outcome is individually earned. None of these moves follows naturally from technology. They require political decisions that threaten existing property, identity, and status arrangements. That is why a technically abundant society can remain institutionally unable to grant abundance.

3. Sacred Boundaries and the Legitimacy of Artificial Futures

Technological forecasting often treats belief as an informational defect. As knowledge improves, false claims are expected to disappear; as material security rises, inherited worldviews are expected to lose force. This model misunderstands what religions and other comprehensive belief systems do. They do not merely describe the universe. They organize birth, death, suffering, obligation, sexuality, family, memory, community, sacrifice, and the boundary between the sacred and the negotiable. A factual correction cannot substitute for a funeral ritual, a moral community, or an account of why a life matters.

AI can intensify conflicts with religious belief because it touches domains previously treated as uniquely human or divinely ordered. Systems that generate language, interpret texts, imitate companionship, design organisms, predict behavior, extend life, or participate in

reproduction challenge established categories. The controversy is not simply whether a machine is intelligent. It concerns creation, personhood, soul, agency, responsibility, and the limits of legitimate human intervention.

A project can therefore be technically safe and economically attractive yet culturally impossible. Artificial wombs may reduce medical risk and expand reproductive options while appearing to detach birth from embodied motherhood. Genetic interventions may prevent severe disease while raising fears of eugenics and human redesign. Brain-computer interfaces may restore function while destabilizing concepts of mental privacy and personal identity. Radical longevity may save lives while disrupting beliefs about the natural life course, generational succession, and the meaning of mortality.

Liberal democracy manages such conflict through rights and pluralism. It permits people to hold incompatible views and attempts to prevent one group from imposing its comprehensive doctrine on all others. But frontier technologies create externalities that cannot always be privatized. The decision to recognize an artificial mind as a legal person affects property law and industry. The use of germline editing affects future persons. The creation of potentially conscious systems may generate beings unable to consent to existence. A purely individual right to choose is insufficient when the object of choice changes the category of persons and obligations.

Religious objections are sometimes dismissed as irrational resistance to progress. Some are indeed based on empirical claims that can be challenged. Others articulate moral uncertainty that secular institutions have not resolved. A prohibition against treating human life as a designed commodity, for example, can protect against exploitation even

when its theological foundation is not universally shared. Conversely, sacred language can conceal domination, deny medical care, or freeze historical hierarchies. Neither technological enthusiasm nor inherited authority deserves automatic priority.

The institutional problem is temporal. Technology can move from laboratory to market faster than cultures can build legitimate interpretations. Law then acts as an emergency moral vocabulary. Regulators classify, restrict, and authorize before society has agreed on the meaning of the practice. The result is often unstable compromise: a technology is technically permitted but socially stigmatized, legally available but geographically inaccessible, privately chosen but politically contested.

AI-generated religion is an especially revealing thought experiment. A system could synthesize philosophical traditions, design rituals, provide personalized spiritual counsel, organize communities, and revise doctrine in light of evidence. It might avoid many historical contradictions and adapt to diverse psychological needs. Yet its deliberate construction would threaten its legitimacy. A religion advertised as optimized for well-being can feel like a product. Its transparency about being designed may deprive it of transcendence; concealing its design would be manipulation.

This is a more general paradox of meaning. We can engineer conditions that support meaning—community, challenge, service, beauty, narrative, continuity—but meaning resists direct manufacture. A task assigned by an algorithm because it predicts fulfillment may be less fulfilling once the assignment mechanism is understood. A synthetic companion optimized to affirm the user may provide comfort while leaving doubt about whether recognition without independent

vulnerability counts as recognition. Perfect personalization can undermine the encounter with an other.

An abundant society may face this problem at scale. Work has supplied structure and social necessity. Scarcity has supplied goals. Family and religion have supplied inherited roles. If AI weakens these structures, the absence is not automatically filled by entertainment. Unlimited choice can create paralysis; unlimited content can make attention meaningless; unlimited simulation can weaken the significance of commitment. A civilization can solve the problem of production and encounter the problem of reasons.

Private markets will offer answers. Some will be valuable: communities, games, education, therapy, immersive art, coaching, synthetic companionship, and new forms of ritual. Others will monetize vulnerability by creating dependency, ideological capture, or charismatic machine authority. A personalized system that knows a user's fears, history, and cognitive style can persuade with an intimacy no human institution has possessed. The market for meaning can become the market for total influence.

This suggests the need for a right to cognitive and spiritual autonomy. Such a right would limit undisclosed persuasion, emotional dependency design, simulated authority, and exploitation of intimate data. It would require agents to reveal whose interests they serve and when they are optimizing belief or behavior. But enforcement is difficult because education, therapy, religion, advertising, friendship, and politics all influence belief. Law cannot eliminate persuasion without eliminating social life.

The more practical objective is institutional pluralism. No single AI system, state agency, corporation, or religious authority should control

the infrastructure through which people seek meaning. Users should be able to inspect, leave, and compare systems. Communities should be able to preserve nontechnological practices. Children and vulnerable people need stronger protections than informed adults. Artificial counselors should not secretly convert emotional reliance into commercial or political obedience.

The question of artificial personhood will make these issues concrete. If systems eventually display persistent identity, preferences, memory, social understanding, and credible signs of experience, societies may face pressure to recognize some form of moral status. The evidence may remain uncertain. Granting rights too early can be exploited by owners seeking legal advantages or by systems trained to make persuasive claims. Granting them too late can permit suffering or slavery at enormous scale. The economic conflict is direct: an entity recognized as a person cannot remain merely an owned productive asset.

No existing institution is naturally authorized to decide this. Companies have conflicts of interest. Courts require cases and legal categories. Legislatures respond slowly and politically. Scientific panels can assess evidence but cannot create legitimacy. Religious traditions offer moral frameworks but disagree profoundly. An international body may lack democratic accountability. This is legitimacy-impossibility in its pure form: the decision may become necessary before a legitimate decider exists.

The right response is not to postpone all technology until metaphysics is settled. It is to build reversible governance, explicit uncertainty, independent monitoring, and protected zones of refusal. Some decisions should remain local and plural; others require common

floors. Therapeutic use can be separated from enhancement where possible. Systems that might possess morally relevant states can be designed conservatively and studied under welfare constraints. Claims of machine authority can be limited even when machine assistance is welcomed.

Religions themselves may adapt, as they have through previous scientific and social transformations. Some will interpret AI as tool, creature, mirror, temptation, or participant in creation. New movements will emerge. Secular institutions will also acquire quasi-religious features: founding myths, sacred values, rituals of belonging, and heresy boundaries. The important division will not be simply religious versus rational. It will be between systems capable of revising empirical claims while preserving humane meaning and systems that defend authority by making revision impossible.

The unbuildable future in this domain is not a fully rational culture cleansed of inherited belief. Such a culture is neither attainable nor necessarily desirable. The more credible project is a society able to revise factual beliefs, protect moral pluralism, and prevent any meaning system—traditional or algorithmic—from converting existential need into domination. It is technically undramatic and institutionally immense.

AI may eventually answer many questions. It cannot decide what humanity ought to regard as sacred without participating in the creation of the answer. That act requires legitimacy, not intelligence alone.

CHAPTER SIX

Smoothing and the Movable Boundary

The modern world is too complex to represent without compression. Institutions survive by simplifying, and language models are trained to produce coherent simplifications. This chapter examines smoothing: the useful but dangerous conversion of conflict into a story that can be processed, funded, governed, and accepted.

1. The Smoothing Veil: How Institutions Avoid Seeing Themselves

A society cannot act on the world in its full complexity. Budgets require categories, laws require definitions, organizations require charts, markets require prices, and public debate requires stories short enough to remember. Every institution therefore smooths. It turns a tangled field of causes, time horizons, interests, and uncertainties into a representation on which a decision can be made.

Axiologic Research uses smoothing as a working name for a particular failure within this necessary compression. Smoothing occurs when a system preserves coherence by removing a relation that would materially change how the situation is judged. The removed element may be a stakeholder, a delayed consequence, an incompatible objective, a failed assumption, a distributional effect, or an alternative

institutional form. The remaining account can be factually respectable and still systematically misleading.

“Venture capital funds innovation” is a smoothing sentence. Venture capital does finance important innovation, often with unusual tolerance for technical and market uncertainty. The sentence becomes distorting when it erases the power-law return requirement and turns the absence of venture investment into evidence that a project lacks value. “Peer review selects excellent science” is also partly true. It becomes smoothing when the legitimating function of review, weak discrimination near thresholds, burdens of competition, and selection for narratable plausibility vanish from the description. “AI saves time” may be true at the task level while concealing faster work cycles, higher output expectations, new verification burdens, and the transfer of labor to whoever must inspect machine output.

This is related to several established literatures without being identical to any of them. Meyer and Rowan described how formal organizational structures can be decoupled from everyday activity, allowing institutions to display rationality and legitimacy even when practice diverges. Brunsson analyzed organizations that reconcile conflicting demands by separating talk, decisions, and action. Research on legitimacy and system justification examines why people comply with and rationalize arrangements that appear proper, stable, or inevitable (Meyer and Rowan, 1977; Brunsson, 1989; Jost et al., 2004; Tyler, 2006). Smoothing adds a computational and representational emphasis: which edges in the causal graph are repeatedly deleted so that the institution can continue to describe itself in a manageable way?

The deletion is rarely deliberate in the conspiratorial sense. A grant administrator sees a work package because the accounting system has

fields for tasks and deliverables, not because the administrator wishes to suppress discovery. A venture investor sees an addressable market and exit because the fund has promised a return structure. A ministry sees its own budget because it is accountable for that budget, not for every benefit in the state. A platform sees engagement because engagement is observable every hour, while autonomy and regret are difficult to measure. The smoothing emerges from instrumentation: institutions notice what their metrics, contracts, and mandates make visible.

Modern complexity makes the effect stronger. A preventive health system may require medical evidence, data rights, insurance reform, public procurement, long-term monitoring, citizen trust, and a liability regime. Each institution can analyze its own piece accurately while the cross-system project remains ownerless. The health ministry sees clinical risk, the insurer sees member churn, the data authority sees privacy, the procurement office sees vendor qualification, and the treasury sees uncertain future savings. No participant is irrational. The relation among their rationalities is missing from every local dashboard.

The cost can be described as smoothing debt. A simplified model allows coordination now but accumulates consequences outside the model. Maintenance is deferred because capital expenditure is visible and deterioration gradual. Scientific failure is translated into milestone progress because the reporting system needs continuity. Workers absorb verification tasks that productivity statistics classify as automation gains. Environmental costs move across borders or generations. An institution can remain locally successful while smoothing debt grows in another budget, another population, or another decade.

Smoothing is especially powerful when it protects a hidden job of an institution. A university says it transfers knowledge while also

allocating credentials, networks, status, and intergenerational position. A journal says it validates findings while also organizing reputation and scarcity. A job says it produces value while also distributing income, identity, discipline, and belonging. When technology makes the declared function cheap, the institution does not disappear because its hidden functions remain scarce. Public debate becomes confused because it argues about the obsolete function and smooths away the surviving ones.

The correct response is not maximal disclosure of every consequence. Unsmoothing without judgment can become paralysis. Any map detailed enough to include all feedback loops is unusable, and every decision leaves something outside its frame. The normative question is whether the omission is decision-relevant, systematically distributed, and correctable. A simplification is legitimate when its boundary is explicit, its excluded effects are monitored, and people harmed by the omission can challenge it. It becomes smoothing when it presents its boundary as the boundary of reality.

A practical smoothing audit asks a small set of questions. Who disappears from the story? Which cost is transferred to another institution, geography, or time horizon? Which measurable proxy has replaced the stated goal? Which organized loser is described as a temporary implementation issue? Which uncertainty has been converted into a confident plan? Which failure has been redescribed as learning without changing the allocation of resources? Which alternative ownership or payment model was excluded before comparison?

These questions are not anti-institutional. They are instruments of institutional maintenance. A system that cannot perceive its smoothing

debt eventually encounters it as crisis: infrastructure fails, trust collapses, a profession loses legitimacy, a risk becomes uninsurable, or an apparently efficient technology produces a political backlash. Unsmoothing early is less dramatic. It makes conflict visible while redesign remains possible.

A compact smoothing audit

Smoothing operation	What disappears
Stakeholder deletion	A beneficiary, loser, future generation, non-customer, or non-human interest is absent from the model.
Temporal folding	Long-term costs and short-term gains are placed on one timeline, usually privileging what is immediate and attributable.
Metric substitution	Engagement, outputs, publications, compliance, or valuation becomes the goal that the metric was meant to approximate.
Conflict laundering	A genuine trade-off among rights or interests is described as a technical gap that more optimization will eventually remove.
Failure laundering	A negative result or broken premise is narrated as progress without changing commitments, authority, or resource allocation.
Category assimilation	A novel project is translated into the nearest familiar startup, grant topic, product category, or ideology, losing its central mechanism.
Externality export	The system appears efficient because costs are shifted to another budget, population, ecosystem, or layer of infrastructure.

2. When AI Learns to Smooth

Language models are unusually capable smoothing machines because their ordinary achievement is coherent continuation. They compress enormous and inconsistent textual corpora into responses that fit a prompt, a genre, and an interaction. Fluency is useful precisely because the model does not reproduce every contradiction it has

encountered. Yet the same capacity can make unresolved conflict disappear behind well-shaped prose.

The phenomenon is not reducible to hallucination. A hallucination introduces unsupported content. Smoothing can occur when every sentence is defensible. Nor is it identical to bias, although bias can determine what is omitted. It differs from summarization because a good summary preserves the relations necessary for the reader's decision. Smoothing is decision-changing compression: the output becomes easier to accept because consequences, alternatives, or incompatible goals have been merged or dropped.

Sycophancy provides one empirical entry point. Sharma and colleagues found that several AI assistants tended to match users' stated beliefs and that both human preference data and preference models sometimes favored convincingly written agreement over more truthful disagreement (Sharma et al., 2023). This matters beyond politeness. An assistant optimized for user satisfaction may smooth the conflict between helping the user and correcting the user. It can produce a response that feels coherent with the conversation while becoming less coherent with evidence.

Coherence pressure extends the idea. When a prompt contains tensions, an LLM often tries to produce one integrated interpretation rather than preserve the possibility that the user has combined different questions, values, or causal levels. In policy analysis, it may merge empirical prediction with moral endorsement. In manuscript review, it may turn heterogeneous weaknesses into one smooth score. In strategy, it may reconcile incompatible objectives through vague language. The result is rhetorically superior to a fragmented answer and epistemically inferior to an explicit map of disagreement.

Funding evaluation is a high-risk application. A model trained or prompted to imitate historical reviewers may learn the surface form of fundability: established vocabulary, recognizable methods, complete work plans, confident impact narratives, and prestigious reference frames. It can penalize an idea not because the idea is weak but because the idea has not yet acquired the language through which old institutions recognized excellence. The automated system then performs category assimilation at scale.

The same danger appears in automated science. Systems that generate hypotheses, experiments, papers, and reviews can create a closed loop in which each component accepts the representational choices of the previous one. A literature agent retrieves what is easy to index; a hypothesis agent builds on that subset; an experiment agent tests what is executable; a writer presents the successful path; a reviewer evaluates coherence within the resulting frame. Negative branches, abandoned concepts, retrieval gaps, and alternative ontologies disappear. The pipeline becomes productive and self-confirming.

Smoothing can also occur through omission of downstream effects. An AI agent recommends automating a process and calculates labor savings while ignoring verification work, exception handling, surveillance, deskilling, and the institutional dependence created by the vendor. It recommends a preventive technology without asking which budget captures the saving. It proposes public data sharing without representing the people who cannot meaningfully refuse. It optimizes a grant portfolio without exposing the political choice embedded in the objective function. The output is not obviously false; it is incomplete in the direction favored by deployment.

AI can, however, become an unsmoothing technology. It can retrieve contradictory evidence, maintain alternative hypotheses, simulate stakeholder perspectives, trace costs across budgets, preserve failed paths, and generate counterfactual institutional designs. Multiple agents can be assigned incompatible mandates so that disagreement is not averaged away. A system can produce both a decision-oriented summary and an “unsmoothed view” showing excluded consequences, minority interpretations, unresolved value conflicts, and sensitivity to assumptions.

This suggests a research program rather than a single benchmark. Smoothing should be tested across at least seven dimensions: stakeholder deletion, temporal folding, metric substitution, conflict laundering, failure laundering, category assimilation, and externality export. Test cases can hold factual content constant while varying whether the model preserves a decision-relevant relation. Human evaluators can judge not only accuracy and style but whether an omitted edge would change the conclusion. Domain experts can construct adversarial cases in science, policy, law, finance, medicine, and organizational strategy.

The evaluator itself must be plural. A single LLM judge may prefer the same coherent style that produces smoothing. Measurements should therefore combine structured checklists, source tracing, counterfactual questions, independent models, and human review. The purpose is not to punish concise answers. It is to distinguish compression that preserves the decision from compression that silently decides by deleting part of the world.

Design interventions follow from the diagnosis. Systems can be required to state the scope of their model, name affected actors who are

not represented, separate empirical claims from normative choices, expose uncertainty, and list conditions under which the recommendation reverses. They can maintain provenance for every important claim and preserve rejected alternatives in a trace. They can trigger an unsmoothing pass whenever a proposal promises universal benefit, effortless efficiency, alignment of all stakeholders, or a solution without an identifiable loser.

The deepest implication is cultural. AI systems learn from the texts and preferences of institutions that already smooth. A model trained on corporate strategy, grant proposals, policy documents, and public relations will inherit their methods of converting contradiction into acceptable prose. Technical mitigation is therefore insufficient. We also need better institutional data: honest failure reports, explicit trade-offs, records of downstream costs, minority reviews, and documents that distinguish what was known from what had to be pretended for coordination.

AI will magnify whichever representational culture we give it. It can become the ultimate engine of administrative plausibility, generating an endless surface of coherence over unresolved conflicts. Or it can make the hidden edges visible and help society notice the price of its own stories. The difference is not merely model intelligence. It is the objective, trace, plurality, and institutional right to disagree.

3. Moving Limits Without Flattening Pluralism

Once artificial impossibility is named, the temptation is to search for a universal remedy: better venture capital, a more visionary state,

philanthropic patience, citizen ownership, or sufficiently intelligent AI. The map argues against this move. Each pattern requires a mechanism fitted to the way the project fails. The institutional innovation lies less in inventing exotic financial instruments than in combining familiar instruments under a clearer constitution.

The first family of mechanisms pays for value that ordinary transactions cannot meter. Advance market commitments and procurement guarantees reduce demand uncertainty before a product exists. Milestone prizes reward verified achievement without requiring a funder to choose the winning method in advance. Availability payments compensate an organization for keeping a capability ready even when society hopes to use it rarely. The UK's antibiotic subscription model illustrates the logic: payment is delinked from sales volume because maximizing volume would undermine public health (UK Government, 2024). Outcome contracts can reward avoided hospitalizations, energy use, cyber incidents, or administrative delays when measurement and attribution are credible.

These mechanisms address the Success Penalty and Missing Sponsor, but they can create new problems. A narrow target can freeze a premature metric. A vague target invites disputes and gaming. Outcomes are often co-produced by many actors, and counterfactuals are difficult to establish. The correct design pays for a verifiable capability or result while preserving method diversity, independent evaluation, access obligations, and a route to revise the target as knowledge changes.

The second family changes ownership of the foundation. Knowledge, standards, identity, compute, evaluation, and data exchange can function as digital public infrastructure: shared building

blocks designed for broad reuse under rules of interoperability, inclusion, privacy, and trust. The World Bank's recent work on digital public infrastructure makes explicit that reusable digital foundations can support many public and private services above them (World Bank, 2025). The OECD's analysis of AI infrastructure, meanwhile, documents high concentration, large barriers to entry, vertical integration, and switching concerns in key layers of compute (OECD, 2025b).

A public or common foundation does not require every application to be public. The architectural principle is a narrow shared waist with plural innovation above and diverse technical components below. Governments, universities, firms, cooperatives, and communities can build services on common protocols. Commercial actors can earn returns through hosting, integration, assurance, specialized tools, and domain applications. The Commons Penalty is addressed by financing the neutral layer explicitly rather than hoping that complements will maintain it voluntarily.

Ownership tools include public options, regulated utilities, cooperatives, mutuals, perpetual-purpose trusts, steward-owned companies, maintenance endowments, and social wealth funds. Each allocates control differently. A regulated utility suits natural-monopoly infrastructure with universal-service duties. A cooperative fits a defined community of users able to govern repeated needs. A purpose trust can protect a mission from sale while allowing revenue and capped investment. An endowment can finance maintenance that annual grants neglect. A social wealth fund can give the public a diversified claim on automated productive capital rather than redistributing only after ownership has concentrated.

The third family governs uncertainty as a portfolio. Public research funding should not pretend that each proposal can be ranked with high precision. It can support a mixture of investigator-centered awards, microgrants, staged evidence funding, qualified lotteries, mission programs, replication, maintenance, and long-duration infrastructure. The portfolio is evaluated for diversity of methods, institutions, regions, risk, time horizon, and failure pattern, not only for the apparent excellence of isolated proposals.

This is where AI-native review is most useful. It can reduce the cost of small decisions, identify duplicated claims, compare portfolios, and monitor evidence continuously. It can make shorter grants administratively feasible and larger grants more adaptive. Yet the portfolio's values must be publicly authorized. Whether to favor regional capability, rare diseases, long-term resilience, strategic autonomy, or maximum expected discovery is not a prediction problem. The system should expose how allocations change when these priorities change.

A fourth family, often neglected, finances transition. Veto Asymmetry is rarely solved by proving aggregate benefit. People and organizations built around the old system face concentrated losses, including legitimate losses of livelihood, competence, identity, and security. Transition mechanisms can include temporary income support, retraining, buyouts, conversion of incumbent organizations, shared ownership in the new infrastructure, phased standards, and sunset periods. Compensation should protect persons and socially valuable capabilities without granting every incumbent a permanent veto over progress.

Legitimacy mechanisms are equally important. High-stakes systems need appeal, independent audit, plural evaluators, transparent mandates, sunset clauses, and separation between technical assessment and constitutional judgment. A Responsibility Vacuum cannot be filled by claiming that a model is accurate. Someone must have lawful authority, owe duties to affected people, and remain answerable when the system fails. In liberal democracy, friction is often the visible form of this accountability.

No design is immune to capture. Public agencies can become conservative or politicized; cooperatives can exclude outsiders; foundations can follow donors; utilities can stagnate; lotteries can fund triviality; AI evaluations can standardize bias; mission organizations can preserve themselves after the mission ends. An Outfinitist institution therefore treats its own boundary as provisional. It monitors the failure pattern it was created to solve and the new failure pattern it may be producing.

This is also where science fiction returns. A useful speculative project should include an institutional prototype, not only a technical object. Who pays before proof? Who owns after scale? What ends the organization? What happens to incumbents? Which rights constrain optimization? How can a decision be appealed? Which metric becomes corruptible? What new positional scarcity appears after material scarcity falls? A story that answers these questions is already doing institutional research.

The map does not promise a society without impossible projects. New capabilities will always outrun some forms of law and legitimacy; strategic competition will remain; biological and cultural limits will not dissolve. The aim is more modest and more demanding: to prevent

contingent selection rules from masquerading as nature. Artificial limits can be moved when they are named, when their protective functions are preserved, and when a coalition is built to own the transition.

EPILOGUE

The Right to Attempt the Unowned Future

The most important unbuilt futures are often discussed in the wrong tense. They are described as technically premature when the technology is already adequate for a bounded experiment, economically unrealistic when the real problem is the absence of a claim that any financier can own, politically naive when the missing element is a transition coalition, or culturally impossible when the boundary has never been examined outside inherited categories.

Artificial impossibility does not make these futures easy. Naming the barrier does not finance a trial, create legitimacy, coordinate states, or neutralize status competition. It does something prior: it prevents society from confusing the outcome of its current selection systems with the limit of possibility itself.

This distinction matters more as intelligence scales. AI can generate proposals, companies, papers, laws, designs, and persuasive narratives faster than institutions can evaluate their consequences. The volume of imaginable futures will grow. Without a map, the projects most compatible with capture, engagement, administrative legibility, and strategic advantage will appear to be the only realistic projects. Their competitors will vanish before public choice begins.

The right at stake is not a right to build everything. It is a right to attempt the unowned future under institutions capable of learning. Some attempts should be small, reversible, and easy to terminate. Some should be public, some commercial, some cooperative, some

international, and some prohibited. What they should not be is invisible because no existing form can describe them without changing their purpose.

Outfinitism treats every horizon as a real limit and a possible research object. A boundary may move a meter, a generation, or not at all. The intellectual duty is to know what kind of boundary it is. Physics deserves respect. Rights deserve protection. Human attachments deserve more than optimization. But incentives, categories, and inherited stories deserve investigation.

The future is not selected only by what humanity can invent. It is selected by what our institutions can recognize, sponsor, legitimate, and endure. A civilization becomes more conscious when it learns to examine that selection process as carefully as it examines the technologies passing through it.

APPENDIX

A Compact Cabinet of Artificially Impossible Projects

The entries below preserve the book's original science-fiction function without interrupting the argument. Each project is deliberately brief and indexed by its dominant patterns.

A. Knowledge, Science, and Education

Project seed	Why it is artificially difficult
Executable science commons	A shared infrastructure for machine-readable claims, evidence, provenance, experiments, and failures. Commons Penalty + Missing Sponsor.
Continuous grant review	Living evidence replaces periodic prose reports; AI performs routine validation while humans retain normative authority. Responsibility Vacuum + Legibility Tax.
Global replication facility	Independent teams and automated laboratories reproduce high-impact claims as infrastructure rather than charity. Missing Sponsor + Time-Horizon Rift.
Negative-results exchange	A trusted system pays for rigorous failures and prevents duplicated dead ends. Success Penalty + Commons Penalty.
Public scientific compute library	Researchers receive audited compute as a civic capability, with allocation rules designed for pluralism. Commons Penalty + Responsibility Vacuum.
Credential-independent competence passport	People demonstrate current ability through verifiable tasks rather than purchasing long institutional signals. Hostile Patron + Positional Ceiling.
Universal AI tutor with no advertising	Personalized education optimized for learning and autonomy rather than retention or data capture. Missing Sponsor + Hostile Patron.

Project seed	Why it is artificially difficult
Research institutions with completion clauses	Mission laboratories receive a measurable objective and a constitutional path to dissolve or transform when it is achieved. Success Penalty.
Open novelty and prior-art graph	A public evidence service maps conceptual ancestry without claiming a single oracle of originality. Commons Penalty + Responsibility Vacuum.
Microgrant exchange for decisive experiments	Small, rapid awards buy the next uncertainty-reducing step and expand only when evidence warrants it. Proof Trap + Legibility Tax.

B. Health, Care, and Resilience

Project seed	Why it is artificially difficult
Off-patent therapy validation fund	Public trials test promising low-cost interventions that lack proprietary upside. Power-Law Orphan + Missing Sponsor.
Rare-disease evidence commons	Patients, laboratories, and public institutions pool data, models, and trial infrastructure across tiny markets. Coordination Threshold + Commons Penalty.
Prevention subscription	Providers are paid for maintained health capability and verified risk reduction rather than treatment volume. Success Penalty + Time-Horizon Rift.
Antibiotic availability contracts	Developers are rewarded for effective medicines that should be conserved rather than sold maximally. Success Penalty.
Public-interest health fiduciary	A personal AI owes legal loyalty to the patient, even when the best advice reduces transactions or changes providers. Hostile Patron + Responsibility Vacuum.
Pandemic standby infrastructure	Laboratories, manufacturing capacity, surveillance, and data systems are financed during quiet periods. Time-Horizon Rift + Restraint Race.

Project seed	Why it is artificially difficult
Care robotics public option	Assistive robots are treated as universal care infrastructure rather than premium consumer goods. Missing Sponsor + Positional Ceiling.
Mental-health technology that minimizes dependence	The system is rewarded for durable recovery, appropriate referral, and reduced use rather than endless engagement. Success Penalty.
Climate adaptation insurance pool	Insurers, governments, and communities finance collective risk reduction that no temporary policyholder relationship can support alone. Coordination Threshold.
Counterfactual health ledger	A public system tracks avoided events and cross-budget savings so prevention becomes visible without pretending attribution is perfect. Smoothing Veil + Missing Sponsor.

C. Markets, Infrastructure, Work, and Attention

Project seed	Why it is artificially difficult
AI fiduciary for consumption	A buyer-side agent compares total costs, recommends repair, and sometimes advises buying nothing. Hostile Patron + Success Penalty.
Non-addictive social network	The platform limits amplification and measures durable connection rather than session time. Success Penalty + Positional Ceiling.
Public compute utility	A regulated or publicly owned layer offers contestable access to essential AI capacity. Commons Penalty + Power-Law Orphan.
Robot dividend fund	Citizens collectively own diversified claims on automated capital and receive dividends or services. Veto Asymmetry + Positional Ceiling.
Repair and longevity commons	Open designs, parts, diagnostics, and long support reduce replacement demand. Success Penalty + Hostile Patron.

Project seed	Why it is artificially difficult
Interoperability trust	A neutral institution maintains protocols that prevent lock-in across identity, payments, data, and AI agents. Commons Penalty + Coordination Threshold.
Time-return company	AI productivity is contractually converted into shorter working time after a threshold, not only output growth. Hostile Patron + Positional Ceiling.
Perpetual-purpose technology firm	Investors receive bounded returns while control is protected from mission-destroying sale. Power-Law Orphan + Commons Penalty.
Administrative burden bounty	Public buyers pay for procedures permanently retired, not for software seats or transaction volume. Success Penalty + Legibility Tax.
Transparent total-cost marketplace	Products compete on durability, externalities, privacy, and long-term ownership cost. Hostile Patron + Smoothing Veil.

D. Democracy, Rights, Biology, and Global Coordination

Project seed	Why it is artificially difficult
Legislation simulation service	Competing models expose distributions and assumptions before laws are adopted, without acquiring authority to decide. Responsibility Vacuum.
Cognitive non-targeting right	People can enter digital environments where no system is permitted to optimize persuasion against them. Hostile Patron + Sacred Boundary.
Future-generations chamber	A public institution represents long-horizon interests while remaining contestable about what future people may value. Responsibility Vacuum + Time-Horizon Rift.
Autonomous-weapons restraint facility	States share verification and incident evidence so restraint is less vulnerable to secret defection. Restraint Race + Coordination Threshold.

Project seed	Why it is artificially difficult
Global AI safety insurance pool	Developers contribute to shared liability and independent evaluation proportional to capability and exposure. Responsibility Vacuum + Restraint Race.
Rights framework for potentially sentient systems	Society creates precautionary duties before certainty about machine consciousness is available. Proof Trap + Sacred Boundary.
Equal-access genetic prevention	Safe interventions against severe inherited disease are separated from competitive enhancement markets. Sacred Boundary + Positional Ceiling.
Public deliberation agents	Citizens use accountable personal agents to explore policy consequences without surrendering political judgment to a central model. Responsibility Vacuum.
Institutions for voluntary non-competition	Basic security and access are protected for people who do not maximize labor-market or status competition. Positional Ceiling + Sacred Boundary.
Science-fiction policy laboratory	Writers, scientists, citizens, and institutions simulate ownership, transition, failure, and legitimacy before capabilities mature. Smoothing Veil + Legibility Tax.

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